ZOO 403
Animal Behavior

Full PPT (All searchable)
Lecture # 1 to 72

Compiled by: **Atiq Rehman**

Lecture # 1

Animal Behavior: Basics

Animal Behavior: Basics

Animal Behavior

"Sum of responses made by an animal to the external and internal stimuli"

Based on:

- Animal's physiological systems and processes

Animal Behavior: Basics

Ethology

"Scientific study of animal behavior in natural environment"

Animal Behavior: Basics

Examples of Behavior

Simple behaviors:

- Moving towards or away from light
- Blinking of eyes
- Reflex movements

Animal Behavior: Basics

Examples of Behavior

Complex behaviors

- Territory defence
- Courtship and mating
- Liking, disliking, fighting or caring
- Migration and navigation
- Building hives, nests, dens
- Producing voices, songs
- Communicating
- Learning, memorizing

Animal Behavior: Basics

Natural Selection in Evolution of Behavior

- Behaviors are essential for survival and reproduction
- Evolution of complex behavioral patterns is subject to natural selection

Animal Behavior: Basics

Natural Selection in Evolution of Behavior

Natural selection contributed to:

- Anatomical development of nervous system
- Enhance physiological features

Animal Behavior: Basics

Objectives of Ethology

Involves understanding of:

- Stimuli that elicit behavior
- Physiological mechanisms mediating response
- Animal's experiences that influence behavior
- Role of behavior in survival & reproduction
- Evolutionary history of behavior

END

Animal Behavior

Lecture # 2

Scientific Approaches to Study Animal Behavior

Scientific Approaches to Study Animal Behavior

Understanding Animal Behavior

Ethologists developed two complementary approaches:

- Neuroethological approach
- Ethological approach

Scientific Approaches to Study Animal Behavior

Neuroethological Approach

Involves:

- Bringing the animal into laboratory
- Observing its behavior in a simplified set of circumstances

Scientific Approaches to Study Animal Behavior

Simplified Set of Circumstances

In the laboratory:

- There are no predators
- Number and sex of conspecifics are controlled
- There are unusual lights, smells and sounds
- Animal is limited to a relatively small, confined space

Scientific Approaches to Study Animal Behavior

Neuroethological Study of Behavior

This approach focuses on:

- Study of neuronal responses/signals generated during the manifestation of an animal's behavior

Scientific Approaches to Study Animal Behavior

Neuroethological Study of Behavior

- Nervous system of animal is exposed surgically
- Recording from neurons as behavior is going on

Scientific Approaches to Study Animal Behavior

Neuroethological Approach—Significance

- Studies under restricted conditions are useful for obtaining neuronal information about behavior
- Answering physiological queries about behavior

Scientific Approaches to Study Animal Behavior

Neuroethological Approach—Limitations

- Such experiments cannot equate an animal's dealing in their everyday lives in natural environment

Scientific Approaches to Study Animal Behavior

Ethological Approach

Involves:

- Going into the field
- Observe animal's behavior in its natural environment

Scientific Approaches to Study Animal Behavior

END

Limitations of Ethological Approach

- For a physiologist, natural conditions render experimental constraints
- It is impossible to record activity from animal's nervous system that is involved in producing behavior

Animal Behavior

Lecture # 3

Proximate Causes of Behavior

Proximate Causes

- The word 'proximal' comes from the Latin word 'proximus', which means 'close' or 'near'.
- Therefore, proximal influences to our behaviors are influences from our immediate situation.

- Proximate factors may be defined as:

  “Those **external stimuli which initiate or maintain biological processes under most favourable ecological conditions**.”

- Proximate causation explains biological function in terms of immediate physiological or environmental factors.
- Example:
  - ✓ Eating to satisfy hunger is a physiological response
  - ✓ A female animal chooses to mate with a particular male during a mate choice trial. A possible proximate explanation states that one male produced a more intense signal, leading to elevated hormone levels in the female producing copulatory behaviour.

Difference between proximate and ultimate causes

1. Genetic–developmental mechanisms	1. Historical pathways leading to a current behavioral trait
Effects of heredity on behavior	Events occurring over evolution
Development of sensory–motor systems via gene–environment interactions	from the origin of the trait to the present
2. Sensory–motor mechanisms	2. Selective processes shaping the history of a behavioral trait
Nervous systems for the detection of environmental stimuli	Past and current usefulness of the behavior in promoting lifetime reproductive success
Hormone systems for adjusting responsiveness to environmental stimuli	
Skeletal–muscular systems for carrying out responses	

Sources: Holekamp and Sherman [532], Sherman [1046], and Tinbergen [1139]

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1. Genetic-developmental mechanism

Effects of heredity on behavior

Development of sensory-motor systems via gene-evironment interactions

2. Sensory-motor mechanisms

Nervous v for the detection of environmental stimuli

Hormone systems for adjusting responsiveness to environmental stimuli

Skeletal-muscular systems for carrying out responses

1. Historical pathways leading to a current behavioral trait

Events occurring ever evolution from the origin of the trait to the present

2. Selective processes shaping the history of a behavioral trait

Past and current usefulness of the behavior in promoting lifetime reproductive success

Animal Behavior

Lecture # 4

Ultimate Causes of Behavior

Ultimate Causes

We ask questions like:

- Why animal behaves as it does?
- What is the evolutionary cause of behavior?

These questions explains the ultimate cause of behavior.

The ultimate level of causation refers to:

- the behavior's evolutionary significance
- how the behavior enhances reproductive fitness

- Ultimate causation explains traits in terms of evolutionary forces acting on them.

Example:

Female animals often display preferences among male display traits, such as song. An ultimate explanation based on sexual selection states that females who display preferences have more vigorous or more attractive male offspring.

Difference between proximate and ultimate causes

1. Genetic–developmental mechanisms	1. Historical pathways leading to a current behavioral trait
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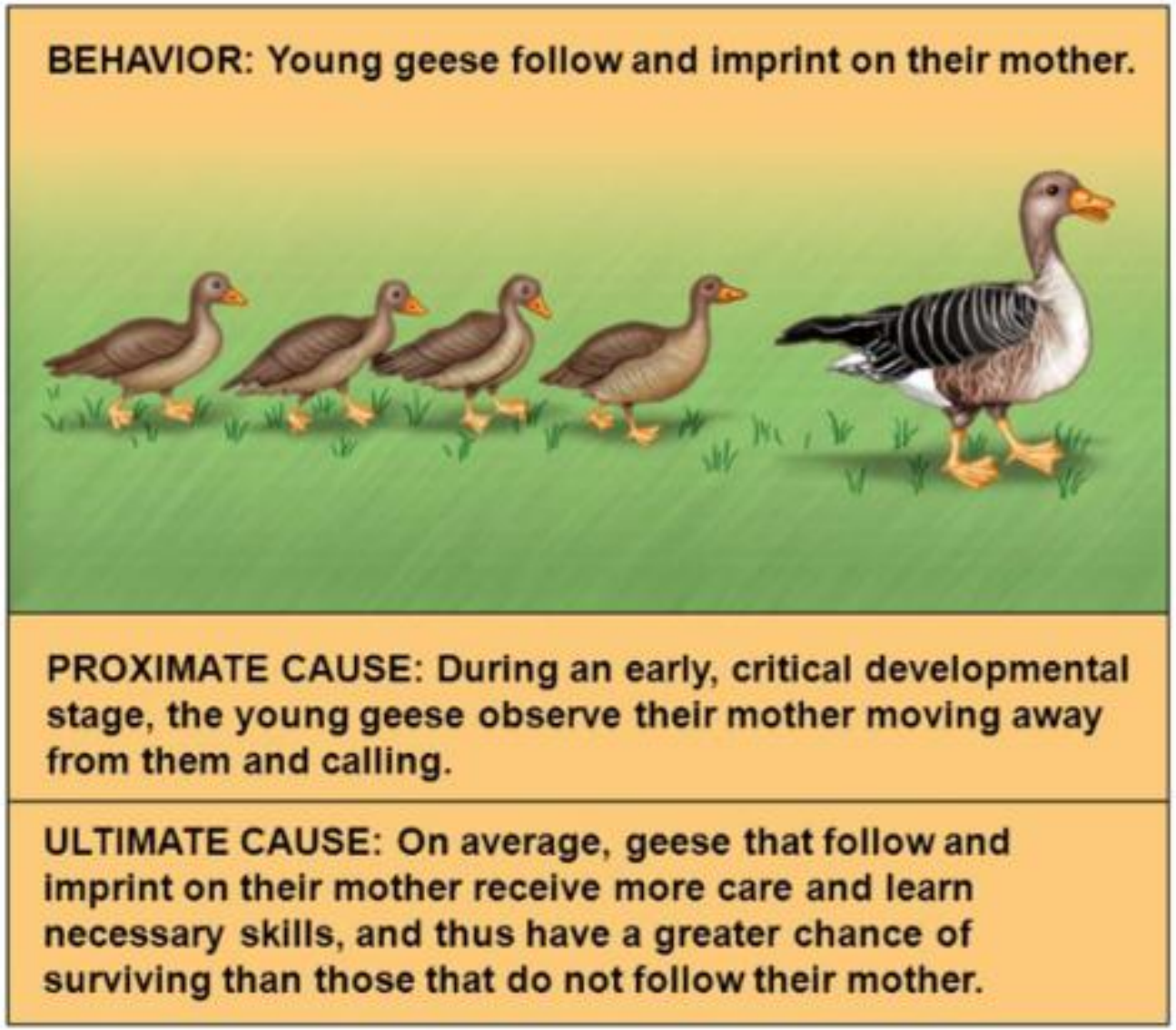
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1. Historical pathways leading to a current behavioral trait

Events occurring ever evolution from the origin of the trait to the present

2. Selective processes shaping the history of a behavioral trait

Past and current usefulness of the behavior in promoting lifetime reproductive success



- BEHAVIOR: Young geese follow and Imprint on their mother.
- PROXIMATE CAUSE: During an early, critical developmental stage, the young geese observe their mother moving away from them and calling.
- ULTIMATE CAUSE: On average, geese that follow and Imprint on their mother receive more care and learn necessary skills, and thus have a greater chance of surviving than those that do not follow their mother.

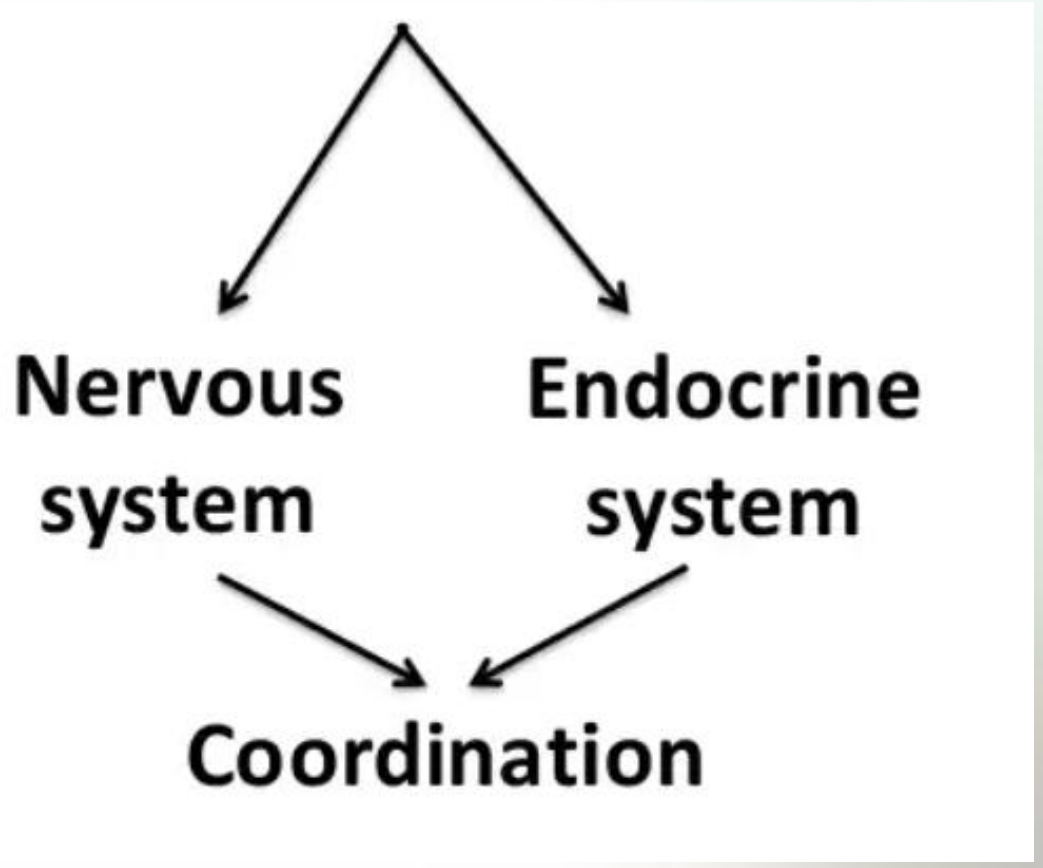
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Animal Behavior

Lecture # 5

Nervous System and Behaviour 1

Behaviour is sum of two systems



Nervous System
Endocrine system
Coordination

Behaviour is sum of two systems

Biologically, behavior is a response to a stimulus shown by the collective mechanism of Nervous and chemical coordinations according to a given situation.

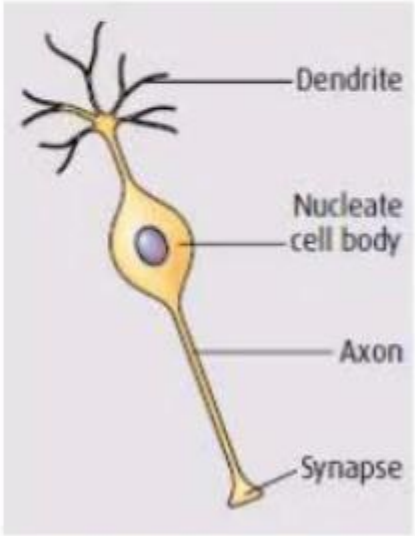
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Nervous system and Behaviour

Components of Nervous system

- Main building blocks and units of nervous system is **"Neuron"**
- Have three components
  1. Cell body
  2. Axon (Move the nerve impulse away from cell body)
  3. Dendron (Move the nerve impulse towards the cell body)



Components of nervous system

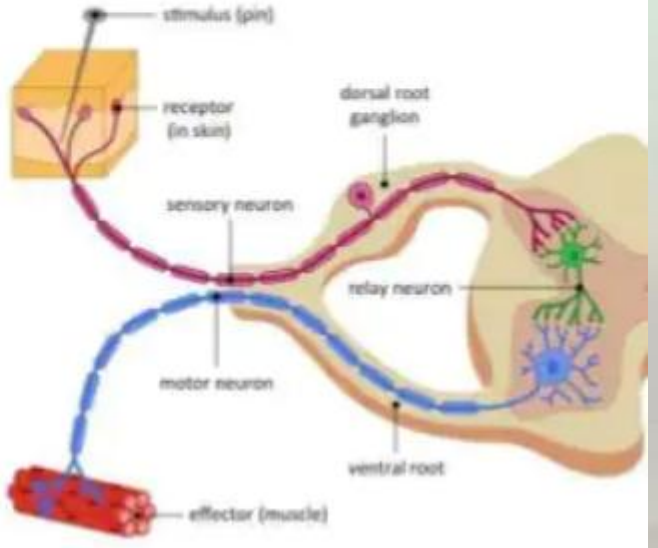
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Nervous system and Behaviour

Classes of neurons

- Three main classes of neurons
  - Sensory neurons
  - Interneurons (relay neuron)
  - Motor neurons
  - "messages through nerve impulse pass in the form of **"Nerve Impulse"**
  - Nerve impulse is an electrochemical change which runs along the length of neuron.



Classes of neurons

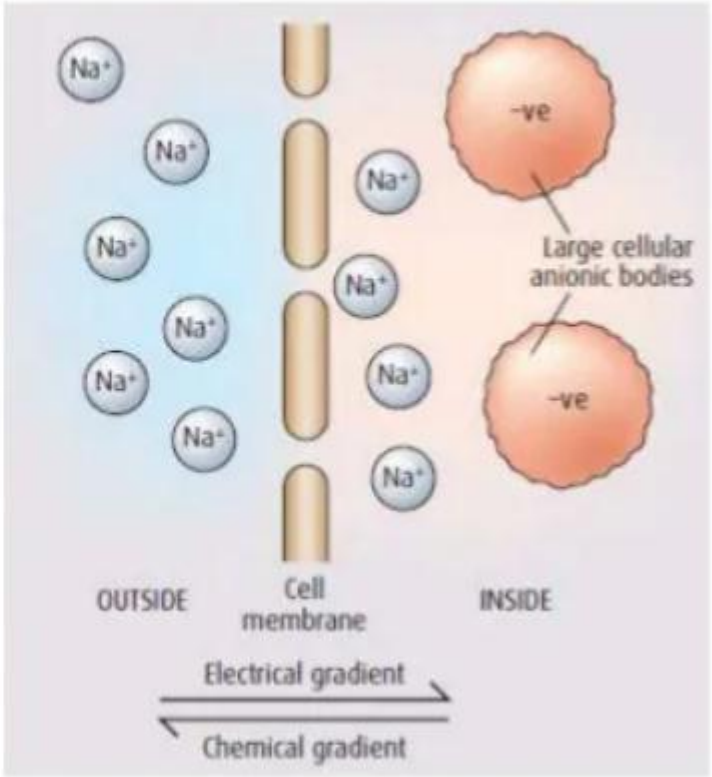
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Nervous system and Behaviour

Nerve Impulse

- The information handled by neurons takes the form of electrical signals.
- These are propagated as small changes in voltage between the inside and outside of the cell.
- To facilitate this the cell membranes of neurons are permeable to ion flow i.e. Na^+ & K^+.



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Nervous system and Behaviour

Membrane Potential: Stored electrical charge of membrane to conduct a nerve impulse.

- Resting membrane potential
  - Two different (Na^+ & K^+) gradients of ion movement across this membrane exist.
  - Ions move down the concentration gradient across the cell membrane, and result in similar gradient one both sides.
  - But membrane is more negatively charged inside due to presence of negatively charged macromolecules i.e. proteins.
  - Na^+, K^+, Ca^{2+} and Cl^- ions have a balanced diffusion gradient across cell membrane known as **equilibrium membrane potential** or **resting membrane potential.**
  - At this stage, membrane is more negatively charged inside than outside i.e. -70mV.

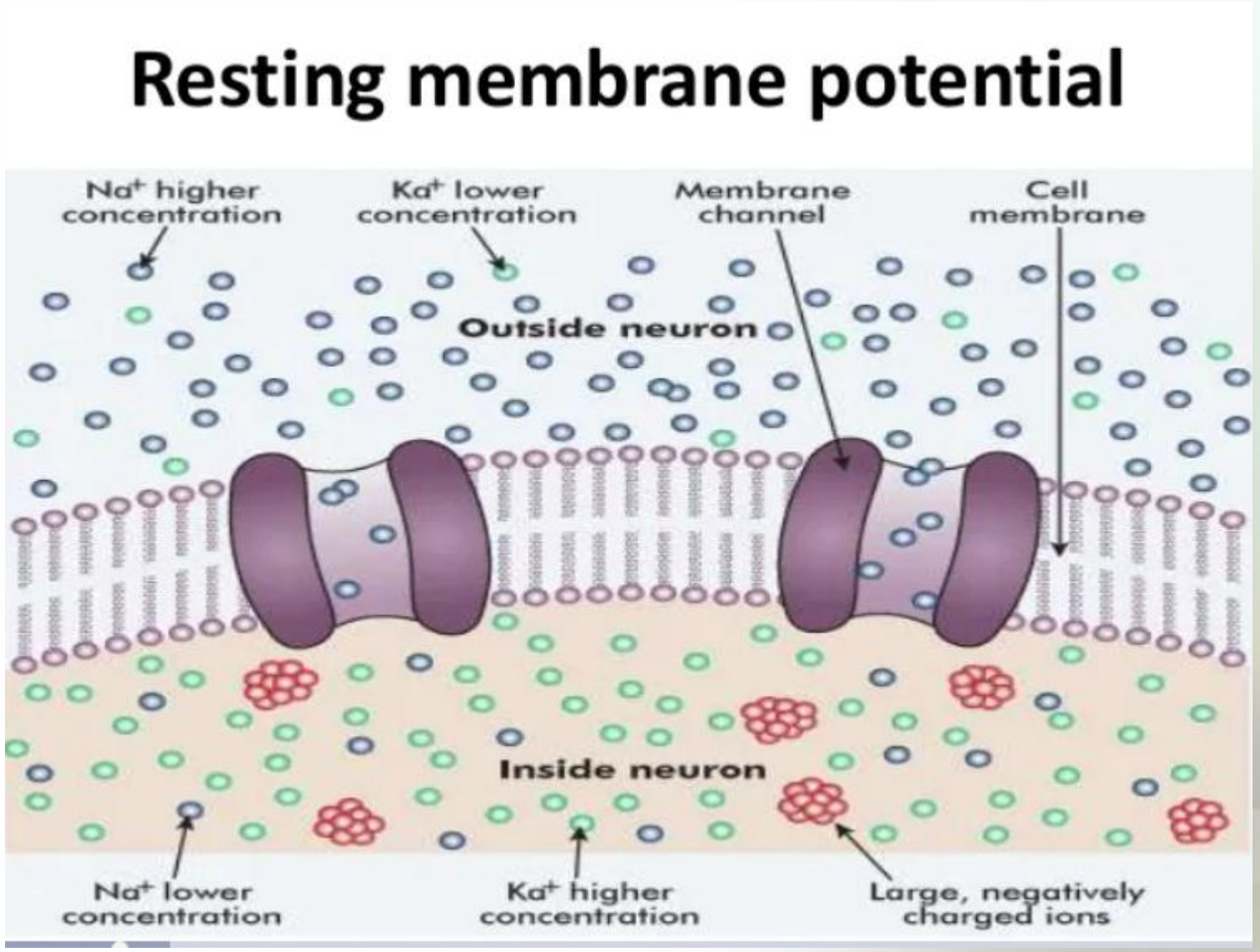
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Resting membrane potential

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Nervous system and Behaviour

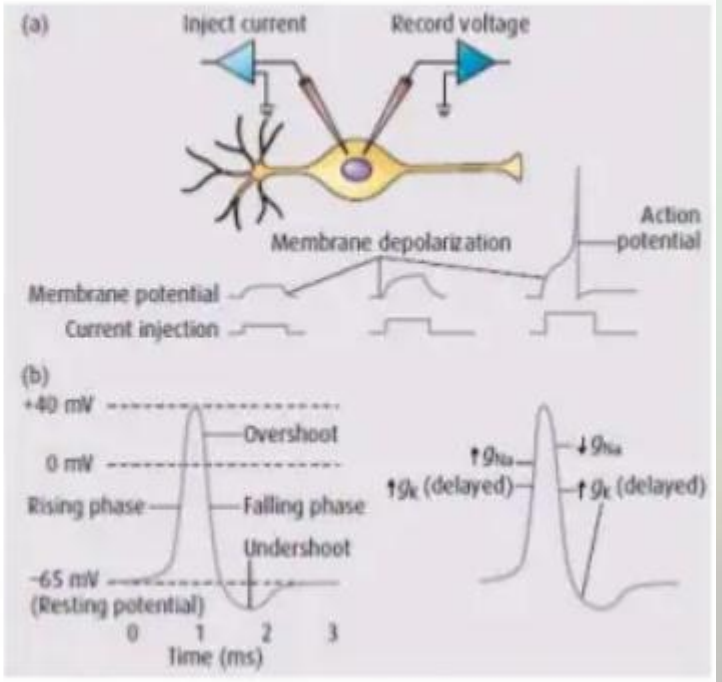


Resting membrane potential

Nervous system and Behaviour

Action Potential

- If the interior of a neuron is made to have a greater positive charge than the resting potential the membrane will depolarize.
- Depolarizations tend to be excitatory in nature, stimulating activity in the neuron, whereby known as **Action Potential**.
- If the charge change related to a depolarization is sufficiently large, a very particular electrical phenomenon called a **spike** will be triggered.
- Action potential is “all or none response”.



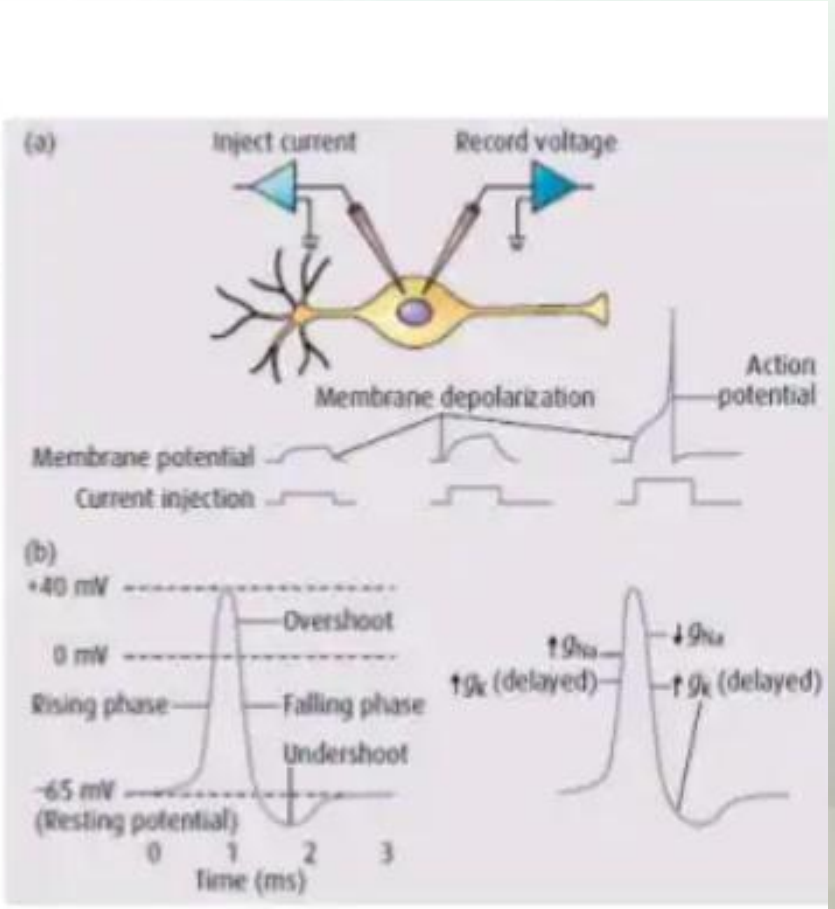
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Hyperpolarization

- The application of an increased negative charge within the cell will result in membrane hyperpolarization.
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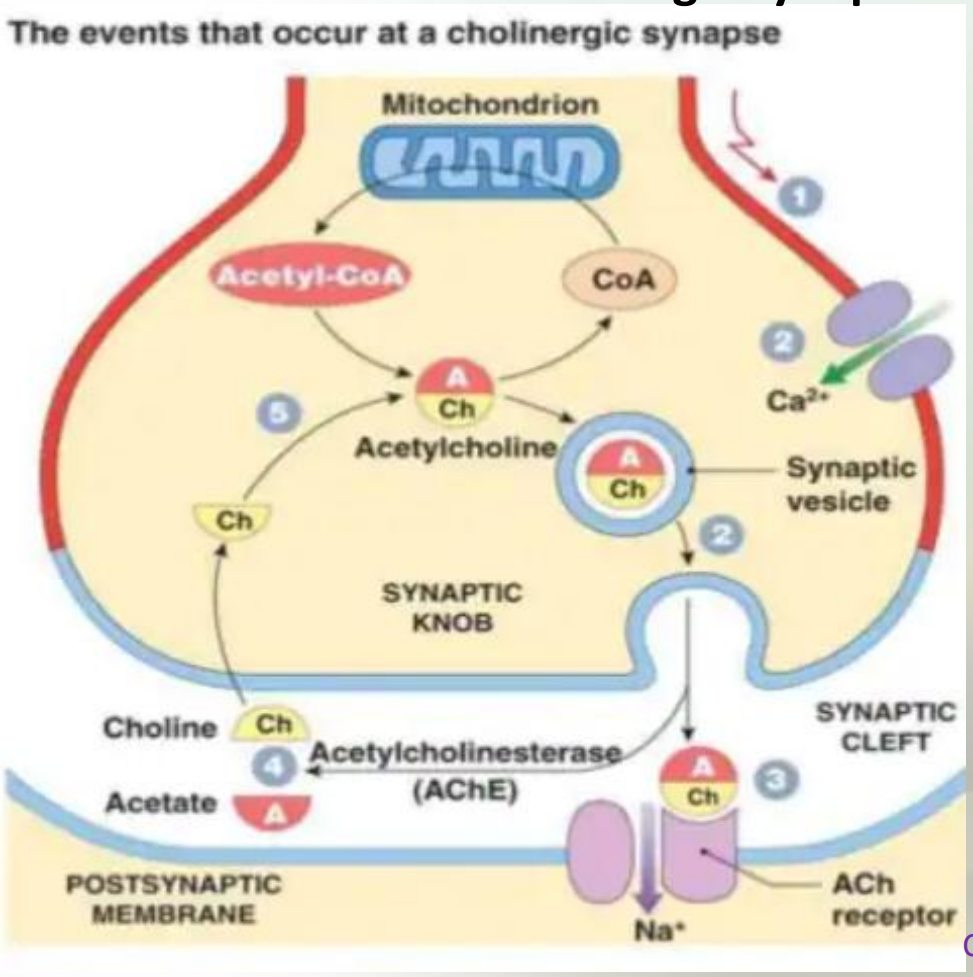
Nervous system and Behaviour

Synapse

Events Occurring at Synapse

1. An arriving action potential depolarizes the synaptic knob.
2. Calcium ions enter the cytoplasm, and after a brief delay, ACh is released through the exocytosis of synaptic vesicles.
3. ACh binds to sodium channel receptors on the postsynaptic membrane, producing a graded depolarization.
4. Depolarization ends as ACh is broken down into acetate and choline by AChE.
5. The synaptic knob reabsorbs choline from the synaptic cleft and uses it to synthesize new molecules of ACh.

Events that Occur at a cholinergic Synapse



Events Occurring at Synapse

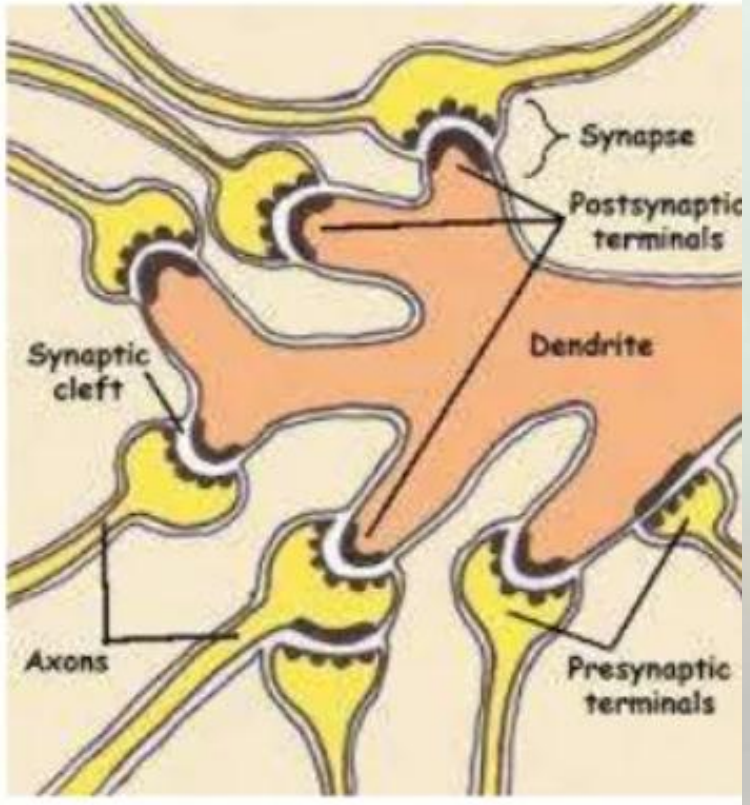
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Nervous system and Behaviour

Potential required to initiate the Impulse

- Sometimes a single postsynaptic potential (PSP) is enough to induce a spike.
- Often the combined effect of a number of them is required.
- Some nerve cells exhibit a phenomenon termed **summation** whereby a series of small PSPs coming from one or a number of sources have an additive effect.



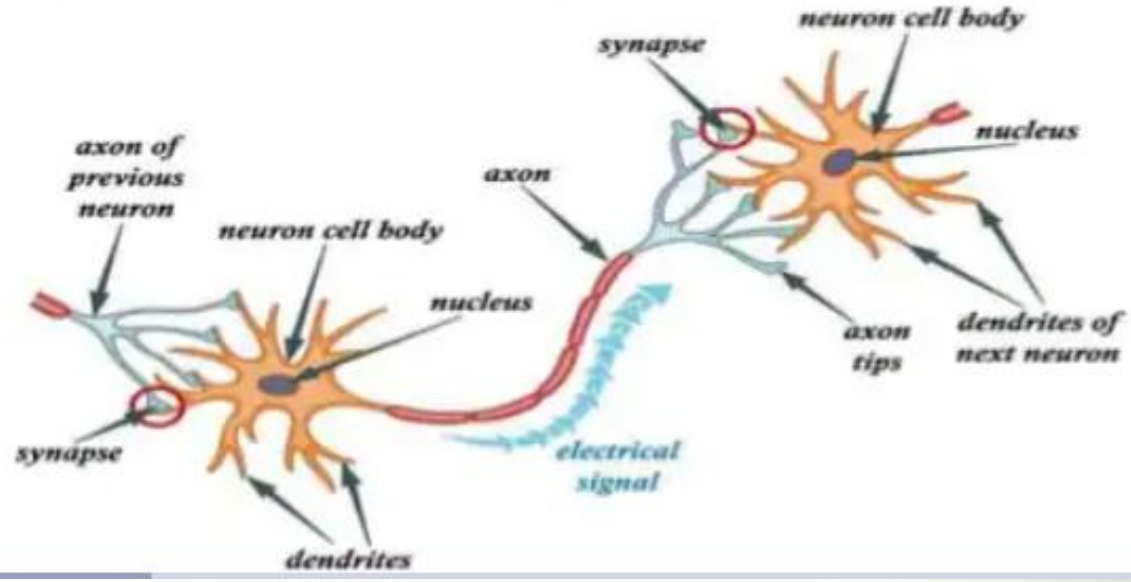
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Nervous system and Behaviour

- Other cells require a series of PSPs that rather than being added together each elicit a slightly larger PSP than themselves until the action potential threshold is reached. This phenomenon is termed **synaptic facilitation.**
- Once a nerve impulse has started, it does not stop until the synapse of the next axon reaches.

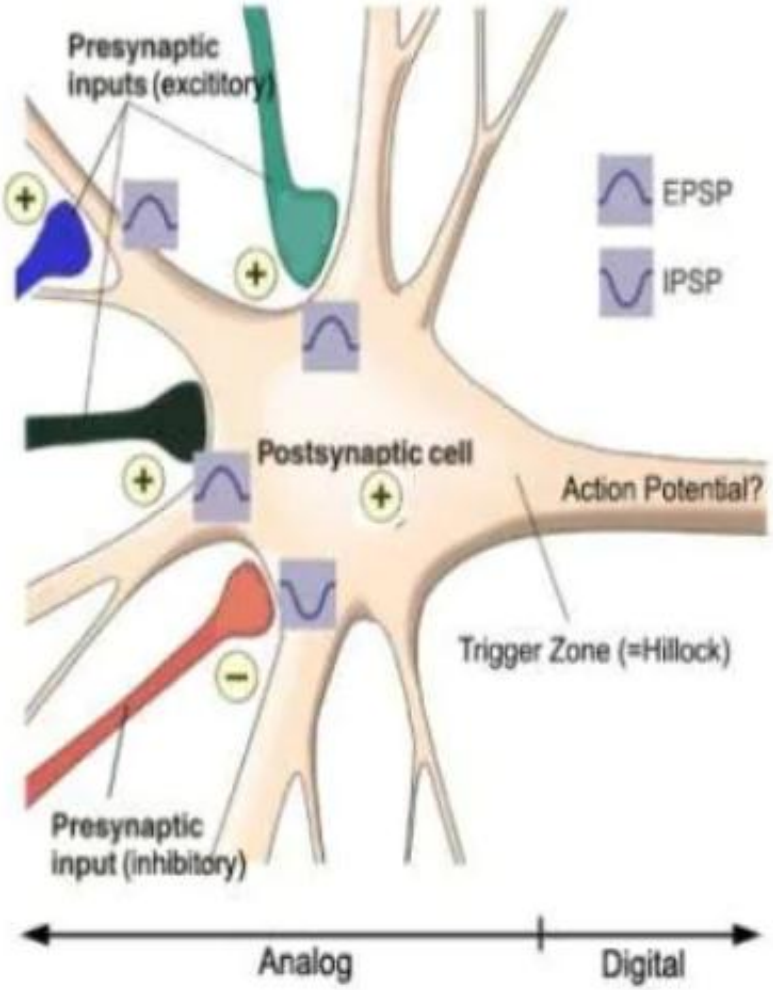


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Nervous system and Behaviour

- The pre-synaptic potential may be Excitatory pre-synaptic potential (**EPSP**) which will depolarize the post synaptic dendron.
- Or it can be Inhibitory pre-synaptic potential (**IPSP**) which will hyperpolarize the post synaptic dendron.



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Nervous system and Behaviour

Integration

- How likely it is that a spike in a presynaptic cell will stimulate a spike in the postsynaptic cell with which it communicates will depend upon the precise combination of EPSPs and IPSPs involved – a process termed **integration**.

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Animal Behavior

Lecture # 6

Nervous System and Behaviour 2 (Examples)

Behaviour is sum of two systems

Examples of Animal Behavior and involvement of Nervous system

Examples of animal behavior and involvement of nervous system

Behaviour is sum of two systems

- To understand the relation of nervous system with animal behavior, we will consider two examples:
  - **Controlling the prey capture (Toad)**
  - **Controlling escape behavior (Cray fish)**

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Nervous system and Behaviour

Controlling the prey capture in toads

Controlling the prey capture in toads

Nervous system and Behaviour

- It was done by Jörg-Peter Ewart and his coworkers.
- They said that **the particular area of nervous system that is responsible for prey capture behavior in toads is called as FEATURE DETECTOR.**
- According to them, during capturing prey, following steps are followed:
  - The toad moves so that it faces the prey animal.
  - The toad approaches the prey animal to within striking distance.
  - The toad uses its tongue to strike at the prey animal and captures it.
  - The toad swallows the animal.
  - The toad wipes its mouth.





Nervous system and Behaviour

- According to them, this behavior of frog is instinctive reflex.
- **Instinct** is any behavior that is innate by its nature i.e. present in an organism by birth.
- They said it is instinctive because according to them"**a newly metamorphosed toad that is completely prey-naïve will perform the whole sequence**"
- Step 1 & 2 can be skipped if the prey is in the very near vicinity.
- So if a toad is treated with a "**model worm**" the nervous coordination of this whole phenomenon can be studied.

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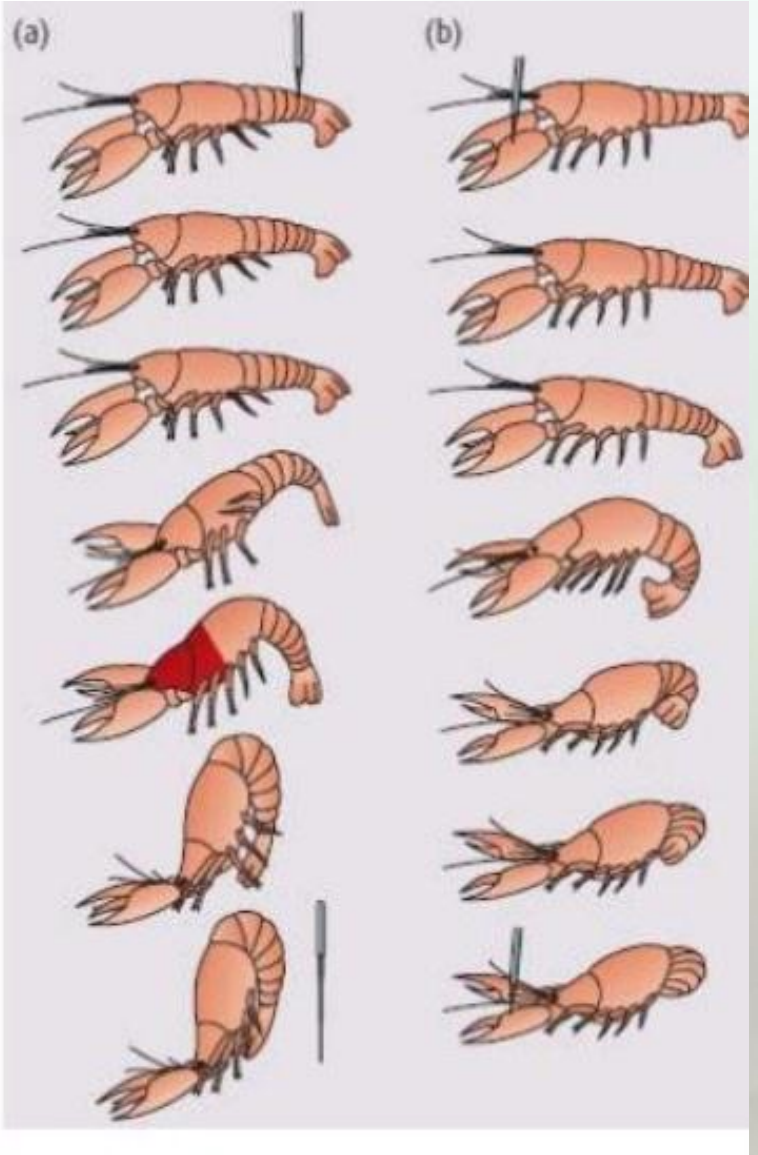
Nervous system and Behaviour

Controlling the escape behavior

Controlling the escape behavior

Nervous system and Behaviour

- The figure illustrates the rapid escape behavior, the tail-flip, of the crayfish.
- Flipping the highly muscular tail propels the animal away from the source of danger, and as one would perhaps expect the onset of the behavior following stimulation is extremely rapid.
- It has a latency (a time period of showing a response) of as little as 10 milliseconds.



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Nervous system and Behaviour

The Nervous Mechanism of tail flip

- Tail flip is caused by rapid contraction and extension of the massive abdominal muscles.
- This activity is controlled by
  - **Large motor neurons** on both sides of the body, coming from Brain.
  - **Motor giants** which sends nerve impulse to every involved muscle fiber.
  - All neurons are interconnected by a number of **Giant Interneurons (GIs)**. GIs are of two types:
    - Medial Giant Interneurons (MGIs): Starting from the brain and all the way into the tail.
    - Lateral giant interneurons (LGIs): Have a very different structure. These interneurons have a chain of cell bodies in every segment.

The Nervous Mechanism of tail flip

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Animal Behavior

Lecture # 7

Neurosensory Basis of Behavior

Neurosensory Basis of Behavior

Role of Nervous System

- NS plays essential role in animal behavior
- Animals manifest behaviors in response to environmental stimuli
- Stimuli act as sensory input to NS
- NS produces motor output
- Motor output controls contraction of muscles which generate all behavioral acts

Neurosensory Basis of Behavior

Neurosensory Basis of Behavior

- Virtually all animal behaviors have neurosensory basis
- Rooted in vast array of neural circuitry (interconnecting circuits of neurons)
- Laying b/w afferent sensory pathways and efferent motor pathways

Neurosensory Basis of Behavior

Neural Circuitry

- Lower animals have simple neural network
- They show simple behavioral patterns
- Neural circuitry is more complex in animals with a CNS

Neurosensory Basis of Behavior

Neural Circuitry

- Increasing number of interneurons in CNS, increases behavioral potential of an animal

Neurosensory Basis of Behavior

Complex Behaviors

- Higher animals with complex CNS show complex behaviors:
  - Learning from experience
  - Memory
  - Associating combinations of stimuli

Neurosensory Basis of Behavior

Complex Behaviors

- Are less predictable—than simple predictable reflex behaviors
- Depend on information stored from past experiences

END

Animal Behavior

Lecture # 8

Neuronal Plasticity

Neuronal Plasticity

Neuronal Plasticity

- “Capacity of nervous system to change with experience”
- Basic architecture of nervous system establishes during embryonic development
- Behavioral patterns modify with time, age and experience
- Premier importance for fitness and survival

Neuronal Plasticity

is the basis of

Examples of Neural Plasticity

- Learning, memory
- Development of motor skills and habits
- Ability to respond to stimuli in diverse ways (different from fixed, programmed reflexes)
- Lies behind human intelligence

Neuronal Plasticity

Mechanism of Neuronal Plasticity

- The reshaping (plasticity) of nervous system occurs at synapses
- Often referred as synaptic plasticity
- Mechanisms underlying synaptic plasticity are currently under intense research

Neuronal Plasticity

involves

Synaptic Plasticity

- Modifying synapses over the course of lifetime due to developmental events
- Synaptic connections established in embryos are refined into adult patterns (mature synapses)
- Synapse maturing results in synaptic strengths enabling learning and memory

Neuronal Plasticity

Synaptic Plasticity

- Changes for learning and memory depend on a retrograde (reverse) signal
- Sent from postsynaptic neuron to presynaptic neuron

Neuronal Plasticity

Mechanisms of Synaptic Plasticity

Occurs by two types of mechanisms:

- Presynaptic mechanisms
- Postsynaptic mechanisms

Neuronal Plasticity

Presynaptic Mechanisms

- Involve changes in the functioning of presynaptic terminals

Example

- A change in the amount of transmitter released from presynaptic terminals

Neuronal Plasticity

Postsynaptic Mechanisms

- Involve changes in postsynaptic neurons

Example

- A change in postsynaptic apparatus resulting in altered frequency of APs

END

Lecture # 9

Physiology of Behaviour in changed Environment 1

Physiology of Behaviour in changed Environment

- In physiology we study the functions and mechanisms in a living system. It focuses on how organisms, organ systems, individual organs, cells, and biomolecules carry out the chemical and physical functions in a living system.

Physiology of Behaviour in changed Environment

- Animals depend on their physical features to help them obtain food, keep safe, build homes, withstand weather, and attract mates.
- These physical features makes it possible for the animal to live in a particular place and in a particular way.

Physiology of Behaviour in changed Environment

Due to the stimulus present animal can show his response in form of:

1. Morphological Responses
2. Behavioral Responses
3. Physiological Responses

Physiology of Behaviour in changed Environment

1. Morphological Responses

- changing in the of the body appearance

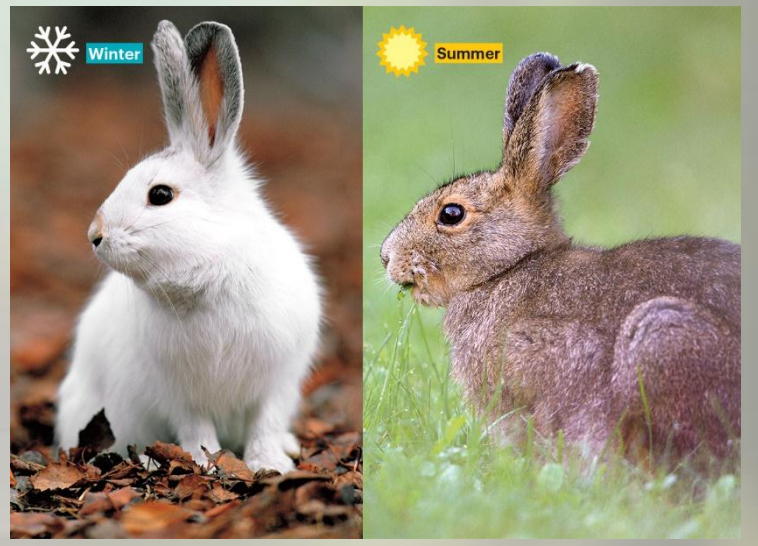
- growing thicker fur or change in fur color in winter

Examples:

- chameleon change colour when it feel threatened.
- Arctic hare change its coat colour from blue-grey to white for better camouflage during winters and turn again to blue-grey in spring.



Chameleon Changing Colour



Arctic Hare

Physiology of Behaviour in changed Environment

2. Behavioral Responses

- changing behavior to adapt to the change
- moving to a more favorable location
- cooperative behavior
- agonistic behavior when threatened

Examples:

- During the winter season, Houbara bustard finds refuge in plain lands of Sindh, Southern Punjab and Balochistan.



Houbara bustard

Examples:

- “Cleaner” fish swim into the mouths of the bigger fish—called “clients”—in order to eat parasites and harmful bacteria. The cleaners get a meal, and the clients get a healthier mouth.





“Cleaner” Fish

Examples:

- In lions, males cooperate to defend the pride, and females hunt and share the kill with other pride members.



Lion’s Pride Hunting

Lecture # 10

Physiology of Behaviour in changed Environment 2

Physiology of Behaviour in changed Environment

3. Physiological Responses

- Changing the functioning of the body
- Acclimation (dilating capillaries to release heat)
- Acclimatization

Physiology of Behaviour in changed Environment



Colour changing in Animals

- Rapid colour change may occur due to various “triggers” including temperature or light (a reflexive response via light-sensitive receptors in skin)
- Most importantly, animals change colour in response to their surroundings including variations in background colour, presence of predators, mates or rivals.

Physiology of Behaviour in changed Environment



Sweating

When the body temperature rises, the sympathetic nervous system stimulates the sweat glands to secrete water to the skin surface, where it cools the body by evaporation. Animals such as horses, monkeys, apes, and hippos sweat etc. use sweating to maintain body temperature.

Physiology of Behaviour in changed Environment



Panting

When an animal pants, it evaporates water from internal body structures. Panting expels hot air and brings in cool air. This physiological response is used by many mammals, most birds, and some reptiles to cope up with the hot environment.

Physiology of Behaviour in changed Environment



Shivering

It is also a physiological response against cold environmental conditions. When an animal shiver its body muscles tighten and relax in rapid succession. This involuntary muscle movement is body's natural response against cold and trying to warm up.

Physiology of Behaviour in changed Environment

Acclimatization

An adaptive change within an individual animal.

Happens due to chronic exposure to new, naturally occurring environmental conditions.

Example

An animal migrates to a high altitude area faces low oxygen partial pressure

Effects

- Lung ventilation rate increases initially
- Ventilation drops to normal in few days
- Animal has acclimatized to the high-altitude

Physiology of Behaviour in changed Environment

Acclimation

Changes are induced experimentally in the laboratory

Example

Simulating high-altitude conditions in lab.

An animal physiologist places the animal in a hypobaric chamber.

Effects

Breathing rate reacts in the same way.

Animal becomes acclimated to the experimental conditions.

Physiology of Behaviour in changed Environment

Both of these are:

- Acquired characters
- Restricted to one or few members of species
- Not inheritable
- Have no evolutionary significance
- Reversible

Lecture # 11

Hormones and Animal Behaviour 1

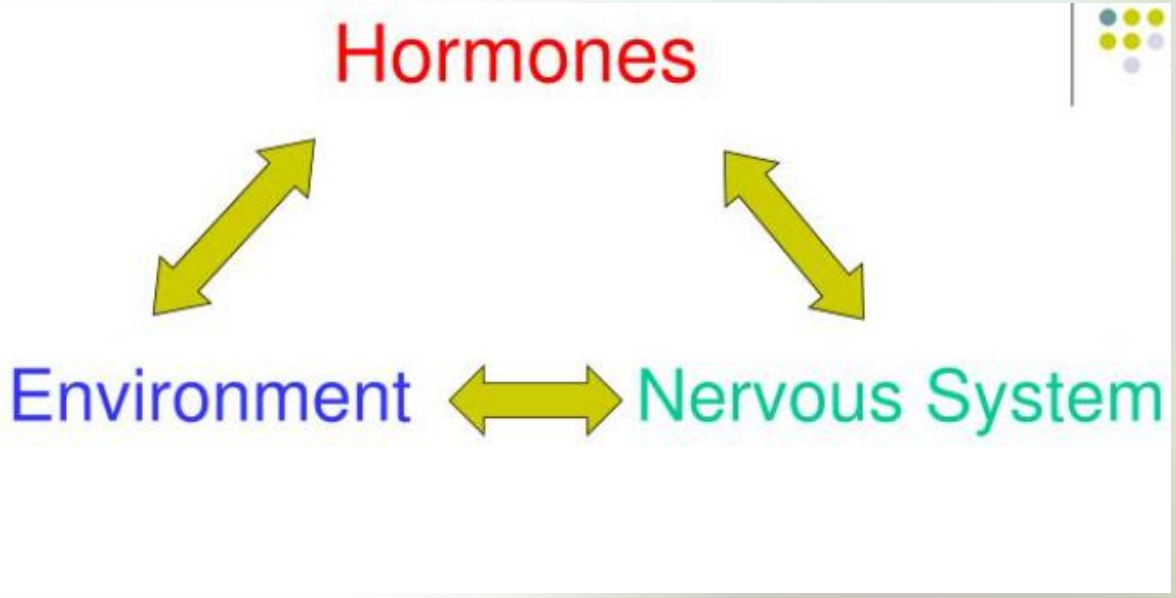
Hormones and Animal Behaviour

The scientific study of the interaction between hormones and behavior is called **behavioral endocrinology.**

Hormones and Animal Behaviour

This interaction is bidirectional: hormones can influence behavior, and behavior can sometimes influence hormone concentrations.

Hormones and Animal Behaviour



Hormone

Environment

Nervous system

Lecture # 12

Hormones and Animal Behaviour 2

Hormones and Animal Behaviour

Hormones are chemical messengers

Released from endocrine glands

Travel through the blood system to influence the nervous system to regulate behaviors such as:

- Aggression
- Mating
- parenting of individuals

Hormones and Animal Behaviour

Examples of hormones that influence behavior:

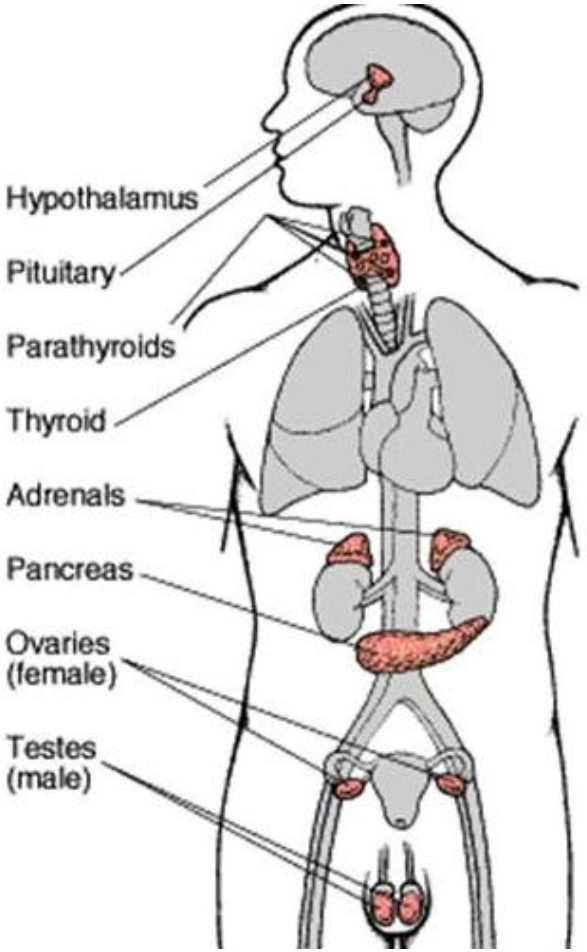
- Steroid hormones such as **testosterone**
- Estradiol (a common type of estrogen)
- Progesterone (a common type of **progestin**)
- Cortisol (a common type of glucocorticoid)

Hormones and Animal Behaviour

Several types of protein or peptide (small protein) hormones also influence behavior, including

- Oxytocin
- Vasopressin
- Prolactin
- Leptin

Hormones and Animal Behaviour



The Endocrine Glands

These are the glands that release the particular hormones.

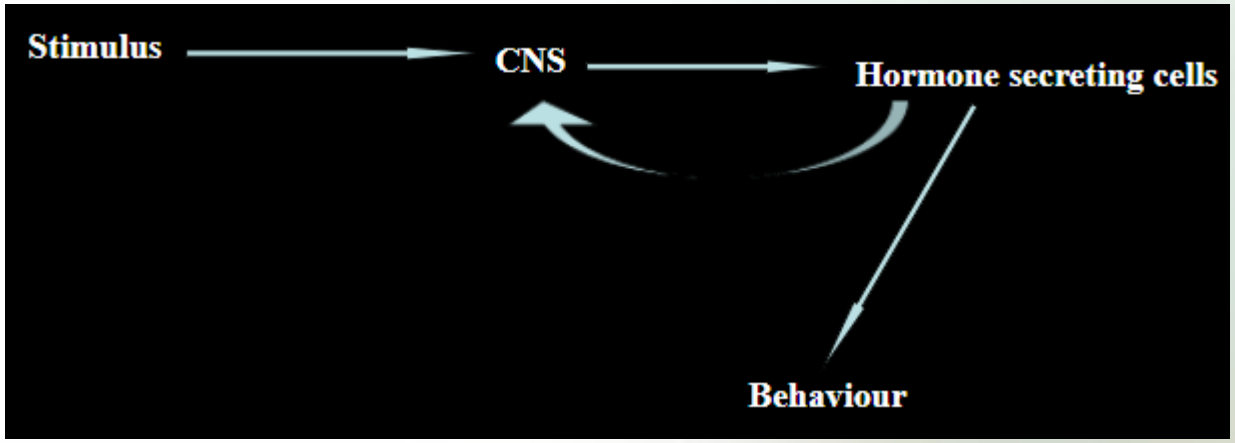
Hormones and Animal Behaviour

Endocrine Glands - What do they do

Gland	Hormone	Effect
Pineal	Melatonin	Annual reproductive cycle
Posterior Pituitary	Oxytocin, vasopressin	Milk ejection, birth, water balance
Anterior pituitary	Luteinizing hormone (LH)	Corpora lutea formation Progesterone secretion Androgen secretion
	Follicle stumulating hormone	Follice development Ovulation (with LH and estrogen.) Spermatogenesis
	Prolactin	Milk secretion Parental behaviour (birds)
	Adrenocorticotropic hormone(ACTH)	Regulates adrenal glands
Intermediate pituitary	Melanophore stimulating hormone	Colour change
Adrenals(cortex)	Steroids	Water balance, metabolic rate, electrolyte balance, blood sugar, stress reaction
Adrenals (medulla)	Adrenaline, noradrenaline	Blood sugar, stress reaction
Pancreas	Insulin	Blood sugar
Testes	Androgens	Testis development, sexual activity, spematogenesis
Ovaries	Estrogens	Uterine growth, mammary gland development, sexual activity
	Progesterones	gestation

Hormones and Animal Behaviour

How effect is brought about



Stimulus

Cns

Hormone secreting cells

behavior

Lecture # 13

Hormones and Animal Behaviour 3

Hormones and Animal Behaviour

Examples of Hormones Control in Animal Behaviour

Anolis Lizards



Testosterone is critical for stimulating male sexual behaviors. Castration reduces the display of male reproductive behaviors, and testosterone can prevent or reverse that effect.

Hormones and Animal Behaviour

- Green Anoles use their pink dewlaps for two purposes.
  - ✓ For breeding, displaying their intent to females.
  - ✓ For establishing territory among adult males. By displaying his dewlap, the lizard can tell other males that they have entered his home range.

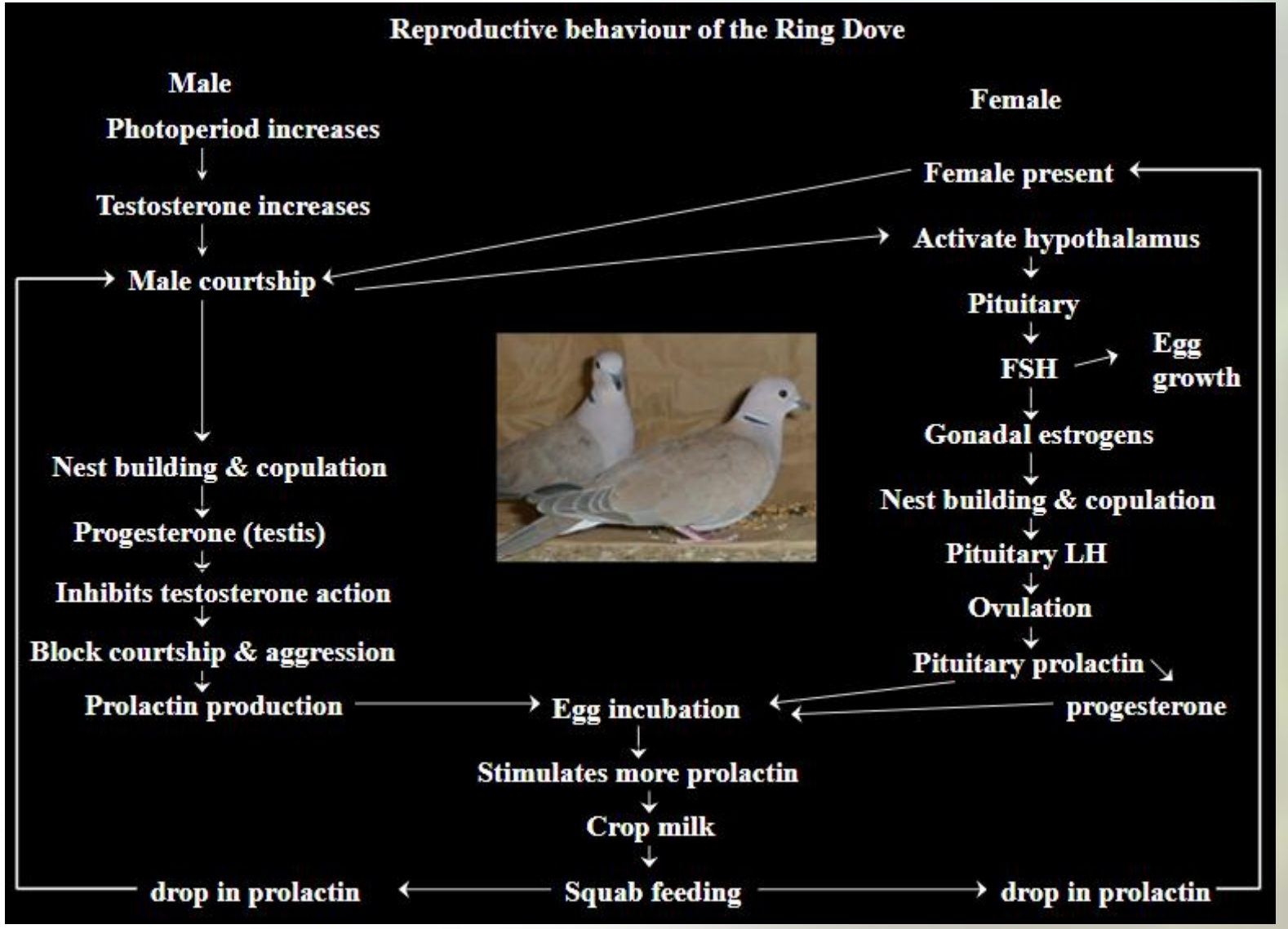
Hormones and Animal Behaviour

Reproductive Behaviour in Ring Doves

Male and female ring doves coordinate their participation in:

- ✓courtship
- ✓nest building
- ✓Incubation
- ✓care of young

Hormones and Animal Behaviour



Animal behaviour

Lecture # 14

Types of Animal behaviour

Types of Animal behaviour

There are 2 basic types of behaviour

1. Innate behaviors or Inherent behaviour
2. Learned behaviors or acquired behaviour

1. Nature/innate

Instinct or genes determine behaviour

2. Nurture/learned

Experience and learning influence behaviour

Innate behaviors or Inherent behaviour

1. **Innate behaviors**
   - automatic, fixed, “built-in”, no “learning curve”
   - despite different environments, all individuals exhibit the behavior
   - Examples: Early survival, reproduction, kinesis, taxis

Innate behaviors or Inherent behaviour

Examples:

1. Web making in spiders



2. Nest building in birds



3. Fighting among male stickleback fish



4. Cocoon spinning in insects



5. Swimming in dolphins



Types of Animal behaviour

2. **Learned behaviors**

- modified by experience
- variable, changeable
- flexible with a complex & changing environment

Types of Animal behaviour

1. By watching their mother, baby ducks learn how to avoid danger and to know what is good to eat. This is an example of learned behavior.



2. Prairie dogs learns by experience to only make alarm call s when they hear predators.



Animal Behavior

Lecture # 15

Inborn, Innate or Instinctive Behaviors

Inborn, Innate or Instinctive Behaviors

- **Innate behaviour** is inborn or innate sequence of activities which are predictable, species specific, genetically controlled and independent of past experience.
- Since this type of behaviour follows a fixed pattern (FAP) which is predictable and found in all members of a species it is also called stereotyped behaviour.

Inborn, Innate or Instinctive Behaviors

Innate or Inborn Behaviors

- Developmentally fixed
- Complex behaviors
- Exhibited by most members of a species
- Are built-in part of nervous system at the time of birth
- Some develop at an appropriate point in maturation

Inborn, Innate or Instinctive Behaviors

Innate or Inborn Behaviors

- Also known as innate, inherent or instinctive behaviors,
- Complex and different from reflexes (simple responses of an organism to a specific stimulus)

Inborn, Innate or Instinctive Behaviors

Characteristics of innate behaviour

- Pattern of behaviour is inherited. It is passed on from parents to offspring.
- It is unlearned behaviour.
- It occurs in all the members of a species hence it is species specific and predictable.
- It is not dependent on past experience as it is an inborn response to a stimulus
- It takes place in individuals even when kept in isolation away from their fellow members
- Innate behaviour has high adaptability and survival value.

Inborn, Innate or Instinctive Behaviors

Genetic Basis of Instinctive Behaviors

- Closely controlled by genes
- Not (or little) influenced by environment

Inborn, Innate or Instinctive Behaviors

Genetic Basis of Instinctive Behaviors

- Instinctive responses are determined by neural programs
- Neural programs are encoded by hereditary material
- Contain information for anatomical and physiological organization of NS

Inborn, Innate or Instinctive Behaviors

Exhibition of Instinctive Behaviors

- Don't have to be learned or practiced
- Are performed in a reasonably complete form even when exhibited first time (without prior exposure or experience)
- Can be modified through experience only to a limited extent

Inborn, Innate or Instinctive Behaviors

Types of innate behaviour

There are different types of innate or stereotyped behaviour

i. Taxis
ii. Irritability
iii. Kinesis
iv. Instincts
v. Reflexes
vi. Motivation

Inborn, Innate or Instinctive Behaviors

Examples of Instinctive Behaviors

- Human newborn’s suckling behavior
- Newly hatched sea turtles, automatically move from beach to the ocean
- A marsupial climbs into its mother's pouch just after birth

Inborn, Innate or Instinctive Behaviors

Examples of Instinctive Behaviors

- Fighting behaviors
- Courtship behaviors
- Building of nests
- Internal escape functions

Inborn, Innate or Instinctive Behaviors

Importance of Innate Behaviors

- Usually involve basic life functions important in survival
- Role varies from species to species
- Especially important for animals with short life spans and poorly developed CNS

Inborn, Innate or Instinctive Behaviors

Role of Instincts in Higher Animals

- Animals with more complex nervous systems rely more on social learning (due to developed cerebral cortex)
- Instincts play lesser role in them

END

Animal Behavior

Lecture # 16

Cocoon spinning in insects such as moths

Examples of Instinctive Behaviors

Cocoon spinning _ an examples of instinctive or innate behaviour

- A cocoon is a protective casing, usually made of silk.
- This casing protects the larval, or immature stage, of an insect from the elements, such as extreme temperatures.
- Inside the cocoon, the larva, which is often wormlike, will undergo metamorphosis and emerge as an adult

Cocoon spinning in insects such as moths

Cocoon-Building Insects

- Butterflies and moths are the most common cocoon building insects.
- Their larvae, which are **caterpillars**, are voracious eaters.
- Caterpillars spin silk, and this silk is used to form the cocoon for the pupal stage of development – the final stage before adulthood.

Cocoon spinning in insects such as moths

Cocoon spinning behaviour

a. Presence of an internal factor pressing for the initiation of spinning.

b. During spinning, turning the entire body around. The spinning behavior includes two distinct and mutually exclusive movement patterns (FAPs)

   i. the stretch-bend

   ii. the swing-swing

Cocoon spinning in insects such as moths

c. Periodically, the silkworm halts one movement pattern, turns Its entire body around, and begins the second series of movements.

d. During period spinning, the silkworm suddenly halts construction of the outer envelope of the cocoon. It then spins a loose, intermediate layer and, finally, a compact inner envelope.

e. If these actions are not elicited by the external environment, they must necessarily be the outcome of events originating within the insect itself.

End

Animal Behavior

Lecture # 17

Web making in spiders

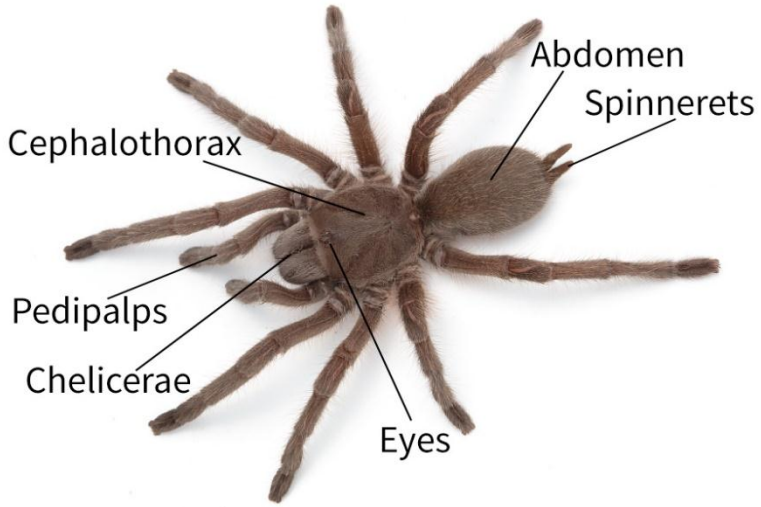
Examples of Instinctive Behaviors



Web making in spiders _ an examples of instinctive or innate behaviour

- A spider web **or cobweb** (from the archaic word coppe, meaning "spider") is a structure created by a spider out of proteinaceous spider silk extruded from its spinnerets, generally meant to catch its prey.

Web making in spiders



- spiders produce silk threads to build their webs.
- silk is produced in silk glands with the help of the spider's spinnerets.
- **Spinnerets** are special organs that allow the spider to decide what type of thread it needs for the web.
- The silk threads can be thick or thin, dry or sticky, beaded or smooth.
- The threads a spider uses to construct its web begin as liquid, but they dry quickly in the air.
- Making webs is instinctive for spiders, which means nobody has to teach them how to do it. They are born knowing how.

Web making in spiders



Process of web making

- When a spider begins a web, it releases a silk thread. It anchors the thread to some object such as a branch.
- As the spider moves back and forth, it adds more threads, strengthening the web and creating a pattern.
- Lines that go from the center of the web outward are called **"radial lines“.** They support the web.
- Threads that go around and around the web are called "**orb lines."**

Web making in spiders

- The main reason spiders spin webs is to catch the prey. When an insect, such as a fly, flies into a spider's web, it gets stuck on the sticky threads.
- When a spider catches prey in the sticky strands of its web, it approaches the trapped insect and uses its fangs to inject venom. The venom either kills or paralyzes the prey.

Animal Behavior

Lecture # 18

Nest building in birds

Examples of Instinctive Behaviors









Nest building in birds_ an examples of instinctive or innate behaviour

- A bird nest is the spot in which a bird lays and incubates its eggs and raises its young.
- For most birds, when a male and female form a breeding pair nest building activities will follow.
- Nest building is an instinctive behaviour as birds have innate ability to build nests.

Nest building in birds

Nest Building Behaviour

- A **nest site** is first selected and nest building begins.
- This can involve both birds as a pair or either sex alone.
- Different species of birds build different nests. Some nests are very simple constructions, whilst others are more complex.

Nest building in birds

- Birds usually only build nests for holding their eggs and rearing chicks.

Birds build nests to:

- To lay their eggs in and keep them safe
- To incubate their eggs
- To raise their chicks
- To protect their eggs and chicks from predators and adverse weather conditions

Nest building in birds

Nests are built in:

- hedges, bushes and trees
- directly on the ground
- Holes, trees or reeds by rivers and lakes
- tunnels in riverbanks and cliffs
- on cliff ledges

Nest building in birds



Weaver Bird's Nest



Crow's Nest



Magpie's Nest

- Sparrow made its nest from stalks of grass.
- The Reed Warbler such as Weaver Bird; weaves its nest and attaches it to reed stems.
- Large birds such as Rooks, Magpies and Crows make very simple flat nests out of twigs. These nests are called platform nests.

Animal Behavior

Lecture # 19

Innate Behaviour of three-spined stickleback fish

Innate Behaviour of three-spined stickleback fish



- Sticklebacks exhibit stereotyped instinctive behaviour such as migration, nest building, territoriality, aggression, courtship and parental care.
- They are a good example of male dominating and nesting behaviour too.
- Normally breed in late March and early August.
- Usually live in schools.
- But at the beginning of breeding season male isolate itself.

Innate Behaviour of three-spined stickleback fish

The reproductive behaviour in three-spined stickleback fish

A sexually mature stickleback fish male will have a red coloured belly to attract the female. A mature stickleback fish migrates to warm shallow waters.

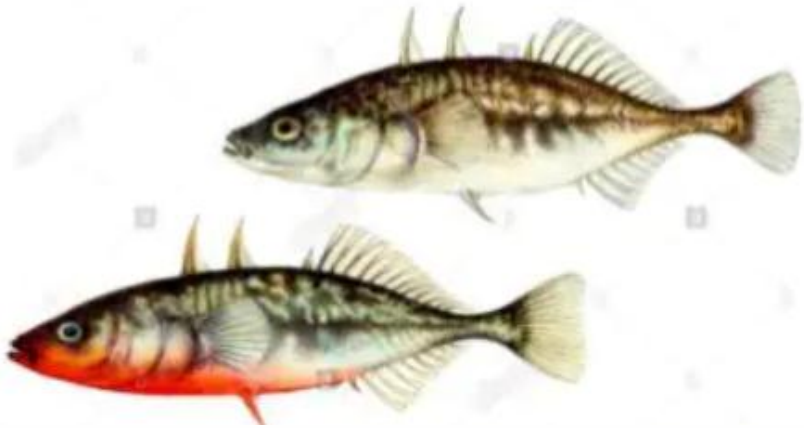
Migration is an instinctive behaviour triggered by the increase in gonadotropic hormones which is in turn controlled by environmental factors like temperature and photoperiod.

Innate Behaviour of three-spined stickleback fish

On reaching shallow waters the fish **builds a nest** which is open at both ends with the help of plants and weeds.

The nest building behaviour is triggered by the presence of green vegetation in the water. Thus he acquires a small territory.

Innate Behaviour of three-spined stickleback fish

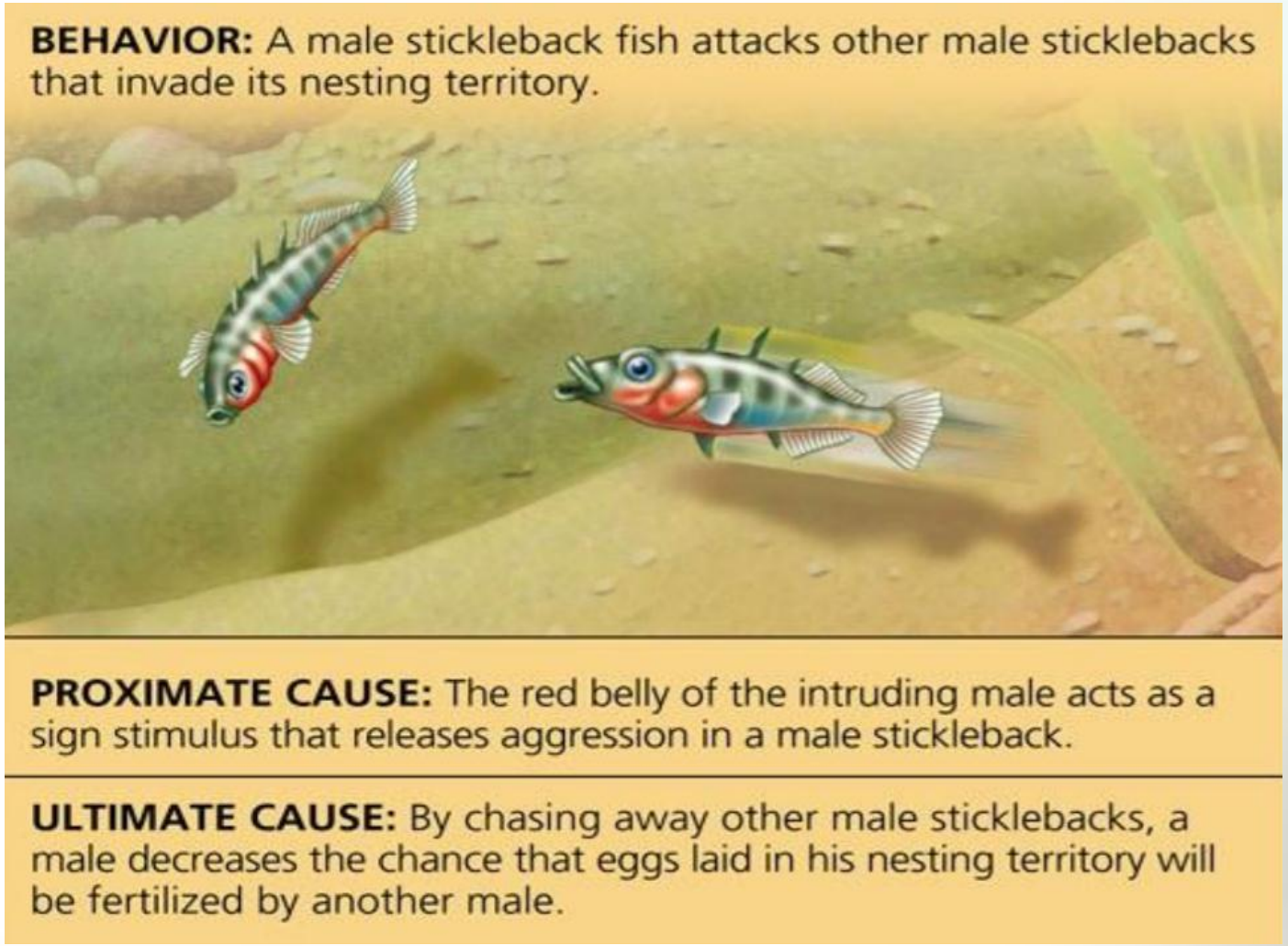


In the early spring.the male stickle backs undergoes quite a dramatic transformation.His throat and belly becomes red-orange,his eyes turns bright blue and silvery scales appears on his back.

It exhibits **territoriality** which means it defends the area around the nest from other male fishes that are also in search of suitable nesting grounds.

The fish will becomes aggressive and exhibits threat posture by facing downwards. The sign stimulus for territorial and aggressive behaviour is the **red belly** of other male fish.

Innate Behaviour of three-spined stickleback fish



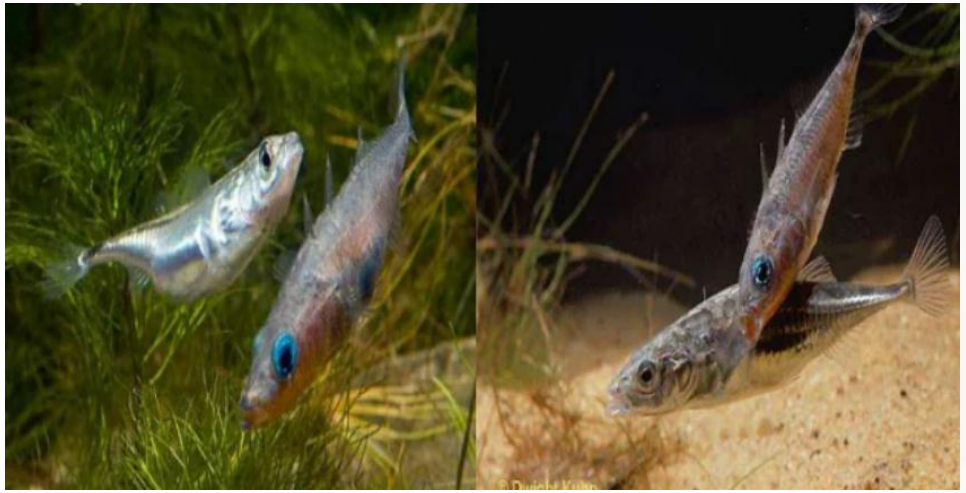
- **BEHAVIOR:** A male stickleback fish attacks other male sticklebacks that invade its nesting territory
- **PROXIMATE CAUSE:** The red belly of the intruding male acts as a sign stimulus that releases aggression in a male stickleback.
- **ULTIMATE CAUSE:** By chasing away other male sticklebacks, a male decreases the chance that eggs laid in his nesting territory will be fertilized by another male.

Compiled by: **Atiq Rehman**

Innate Behaviour of three-spined stickleback fish

The presence of a female triggers the **courtship behaviour** in the male because of its swollen belly that is full of eggs.

The male performs a ziz-zag dance which will attracts the female. During his dance the male always exposes his red belly to the female.





Innate Behaviour of three-spined stickleback fish



The male leads the female into the nest to releases its ova that are immediately fertilized by the male. After this the female is driven out and the male may lure other females to its nest.

The presence of eggs in the nest triggers **parental care in the male fishes** and it will begin to fan the eggs with its fins and young once are taken care by the males.

Innate Behaviour of three-spined stickleback fish

Hence various stereotyped instinctive behaviour in stickleback fishes are:

- Migration
- nest building
- Territoriality
- Aggression
- courtship and
- parental care

Innate Behaviour of three-spined stickleback fish



Animal Behavior

Lecture # 20

Swimming in dolphins and other aquatic species

Swimming in dolphins and other aquatic species



- Dolphins are marine mammals closely related to whales and porpoises.
- There are almost forty species of dolphins.
- Dolphins are among the intelligent animals, and their often friendly appearance and seemingly playful attitude have made them popular in human culture.

Swimming in dolphins and other aquatic species



Physical Features

- They have a pointy snout that makes them look like they are smiling.
- They are carnivores.
- Dolphins have a blowhole at the top of their heads that they use to breathe.
- They have upper and lower jaws with small teeth.
- Dolphins have a dorsal fin which keeps them steady while they are swimming and two tail flukes for moving up and down.

Swimming in dolphins and other aquatic species

HABITAT

Dolphins are found all over the world; generally in shallow cold waters as well as the warm tropical waters

Swimming in dolphins and other aquatic species

BEHAVIOR

- They eat between six and fifteen kg. of food per day.
- They swallow their food whole even though they have so many teeth.
- When bottlenose dolphins are ready to eat, they send out a scout to find the food.
- When dolphins are looking for a mate, it’s usually the female who looks for the male.

Swimming in dolphins and other aquatic species

Behavioral Adaptation

Dolphins work together to find food taking turns to feed while the other dolphins keep the fish herded.

Swimming in dolphins and other aquatic species

Innate Behaviour in Dolphins

Dolphins may have lots of learned behaviour but some of the innate behaviour also present in them:

- Know how to swim by birth
- Leaping from water
- Preying and eating food
- communicate with each other using a wide variety of vocalizations

Swimming in dolphins and other aquatic species

Innate Behaviour in Dolphins

- Observations indicate that dolphins undergo daily cycles of activity, which include feeding, socializing, traveling, and resting. All these are innate or instinctive behaviours of dolphins.
- Bottlenose dolphins show aggression and establish and maintain dominance through posturing, biting, chasing, jaw clapping, smacking their tails on the water, emitting bubble clouds from their blowholes, and impact through body slamming in the most severe cases.

End

Animal Behavior

Lecture # 21

Animal Orientation, Kinesis and Taxis

Animal Orientation, Kinesis and Taxis

Animal Orientation

"Ability of an animal to determine its position in space and move predictably with respect to specific stimuli"

Animal Orientation, Kinesis and Taxis

Animal Orientation: Examples

- Standing up after falling down (orientation against gravity)
- Orienting towards incoming light
- Acoustic orientation
- Olfactory orientation

Animal Orientation, Kinesis and Taxis

Sensory Motor Basis of Orientation

- Animal orientation is a complex process
- Requires integration of sensory input and coordination of motor output

Animal Orientation, Kinesis and Taxis

Input-Output Integration

Integration includes:

- Receiving information through sensory receptor neurons
- Processing and correlating it in the CNS
- Forming a response through muscles that cause orienting movement

Animal Orientation, Kinesis and Taxis

Types of Orienting Movements

- Kinesis
- Taxis

Kinesis

"a change in activity or turning rate in response to a change in location"

Animal Orientation, Kinesis and Taxis

Kinesis—Example

- Kinesis in sow bugs in response to variation in humidity
- Move more rapidly in dry areas and become slow in humid areas
- They survive better in moist areas
- Rapid movement in dry areas to reach moist one is significant for survival

Animal Orientation, Kinesis and Taxis

Taxis

"An oriented movement toward (positive taxis) or away (negative taxis) from some stimulus"

Animal Orientation, Kinesis and Taxis

END

Taxis—Examples

- River fishes (e.g. trout) swim or orient in an upstream direction (toward the current)
- Helps keep the fish from being swept away
- Also keeps facing the direction from which food may come
- Cockroaches move away from light (-ve phototaxis)

Animal Behavior

Lecture # 22

Types of Kinesis

Types of Kinesis

Kinesis

- Kinesis is a type of locomotory behaviour in relation to the source of stimulus.
- The animal responds to the variation in the intensity of the stimulus and not the source of direction of the stimulus.
- To respond to such stimulus the animal only requires sense organs sensitive to variation in stimulus intensity.

Types of Kinesis

There are two types of kinesis

1. Orthokinesis
2. Klinokinesis

Types of Kinesis

Ortho kinesis

It is a response that involves changes in the speed of movement of the whole body in response to stimuli like humidity, preserve and diffused light.

Types of Kinesis



Examples:

- Wood louse, *Porcellerio scaber,* tend to lose water from their body fairly rapidly when exposed to low humidity.
- When kept at the junction in a choice chamber where one side has high humidity and the other half has low humidity.
- It begin to move and the speed of movement and the rate of turning is greatest in the driest part of the chamber.
- This increased and apparently random movement is an attempt by the animal to remain in the most favorable environment.

Types of Kinesis

Klinokinesis

The speed of locomotion remains constant but the rate at which the animal changes direction depends on the intensity of the stimulus.

Types of Kinesis



Examples

- A planaria changes its direction ever so often as it crawls. If the light intensity above the animal is increased it changes direction more frequently but moves at the same speed .
- The increase in rate of turning falls after sometime but increases again with a further increase in the intensity of light.

End

Animal Behavior

Lecture # 23

Taxis and its types

Taxis and its types

Taxis

Taxis is the simplest type of innate or stereotyped behaviour. It is an orientation of an animal (directed either towards or away) in response to the source of stimulus.

Taxis and its types

Types

- **Positive taxis**

If the orientation is towards the stimulus it is called as positive taxis

- **Negative taxis**

If the orientation is away from the stimulus it is known as negative taxis

Taxis and its types

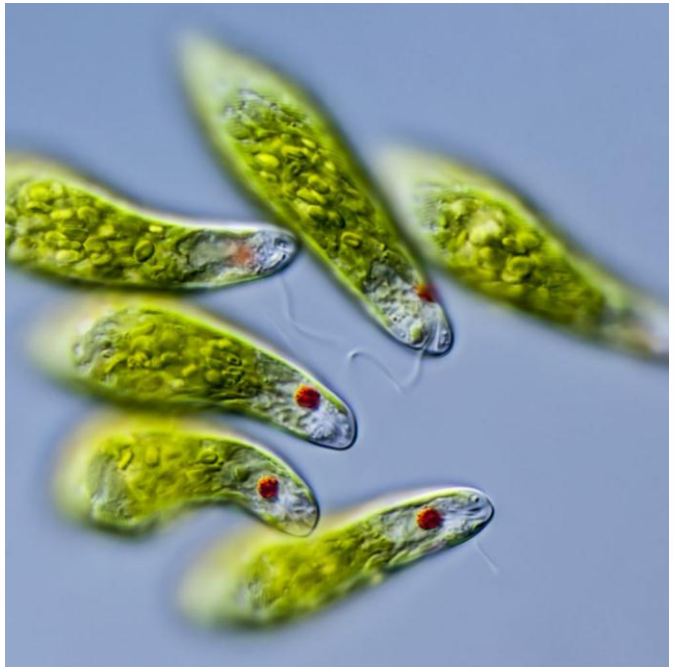
Taxes are usually named after the stimuli. Hence there are:

- Phototaxis
- Chemotaxis
- Thermotaxis
- Geotaxis

Taxis and its types

Examples

- A plant or animal that move towards light shows positive phototaxis.
  - The protozoan, Euglena response to a variety of stimuli and is very sensitive to light. It swims towards light; hence it shows positive phototaxis.



Taxis and its types



End

Examples

- Certain burrowing animals more away from the source of light exhibits negative phototaxis.
  - Maggots that are about to pupate moves away from light source towards a dark location hence they show negative phototaxis.

Animal Behavior

Lecture # 24

Reflexes and its advantages

Reflexes and its advantages

Reflexes

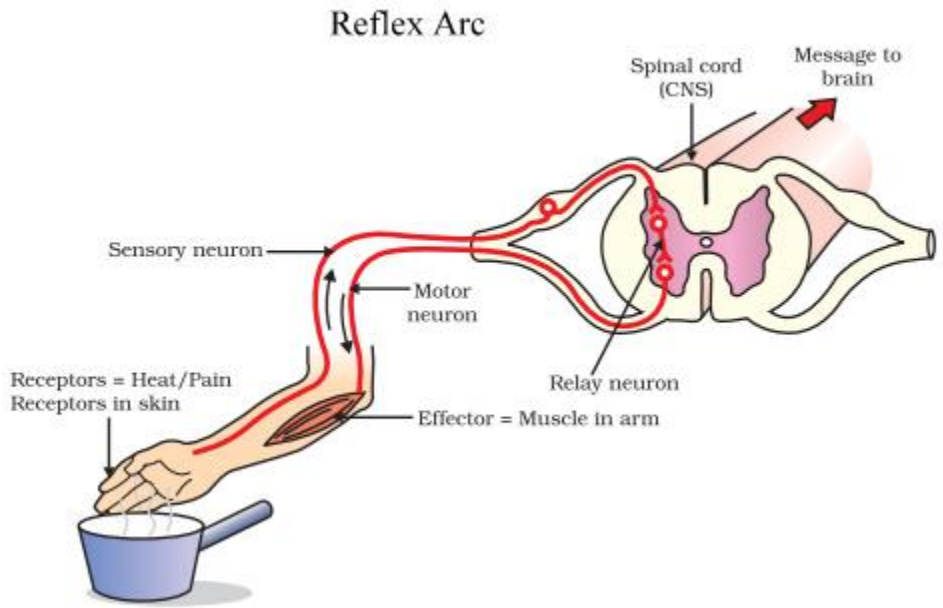
- A simple movement of a part of the animal in response to a stimulus is called **reflex**.
- It is a quick, innate and immediate response of a part of the body to an external or internal stimulus which has great adaptive and survival value to the organism.
- Reflexes are inherited and unlearned behaviour found in all members of the species.

Reflexes and its advantages

Examples

- Kneejerk
- Constrict of pupil of eye in the bright light
- Blinking of eye
- Peristalsis and coughing in man
- Flight in birds
- Web spinning in spiders

Reflexes and its advantages



A **reflex action** requires a **reflex arc** which consists of:

- a sensory organ (receptor),
- a sensory nerve (afferent nerve),
- the spinal cord or brain,
- an intermediate motor or efferent nerve
- a motor organ or effector

Reflexes and its advantages

A reflex action thus involves the sense organs and nervous system.

Steps

1. The stimulus is received by the receptor or sense organ which sets up a sensory impulse.
2. This impulse is transmitted to the central nervous system (brain and spinal cord) through the intermediate nerve.
3. This motor impulse is transmitted to the effector or motor organ, muscles and glands which either contract or secrete in response to the stimulus.

Many pattern of animal behaviour are complex combinations of simple reflex.

Reflexes and its advantages

Advantages of reflex

- Enables the animal to respond immediately to harmful stimuli hence it has great adaptive and survival value
- Since many of the reflex actions are controlled by the spinal cord, it relieves the brain from too much work.

End

Animal Behavior

Lecture # 25

Fixed Action Patterns

Fixed Action Patterns

Fixed Action Patterns (Instincts)

- It is a sequence of unlearned, innate behaviour that is unchangeable.
- Once initiated; it is usually carried to completion.
- It is hardwired, however, not purely genetic, may improve with experience.
- FAP is triggered by an external sensory stimulus known as “sign stimulus”.

Fixed Action Patterns

Konrad Lorenz is the founder of the concept.

Lorenz identified **six characteristics** of fixed action patterns. FAPs are:

- Stereotyped
- Complex
- Species-characteristic
- Released
- Triggered
- Independent of experience

Fixed Action Patterns

Properties of Fixed Action Patterns

- Are relatively complex motor acts
- Are all-or-none responses i.e. once initiated, continue to completion even if the stimulus is removed

Fixed Action Patterns

Properties of Fixed Action Patterns

- All members of a species perform a given fixed action pattern identically
- Patterns are inherited genetically
- Are exhibited perfectly without prior experience

Fixed Action Patterns

Properties of Fixed Action Patterns

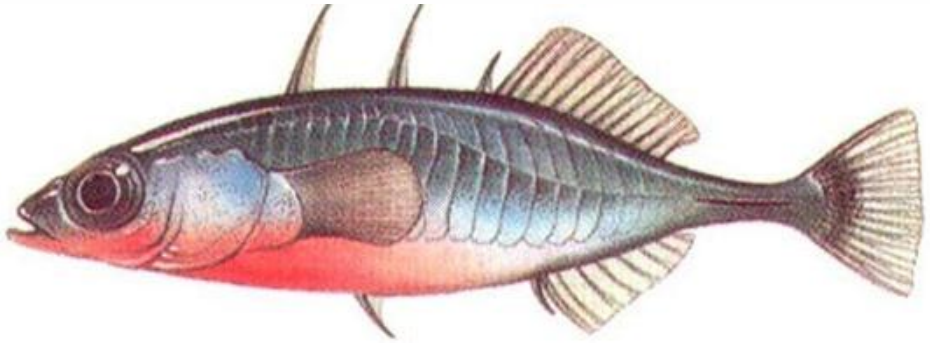
- Stimulus threshold may vary with the physiologic state of animal. For example:
  - An animal has copulated once to an arousing stimulus
  - It cannot be induced to copulate again without intense stimulation

Animal Behavior

Lecture # 26

Fixed Action Patterns 2

Fixed Action Patterns



Example

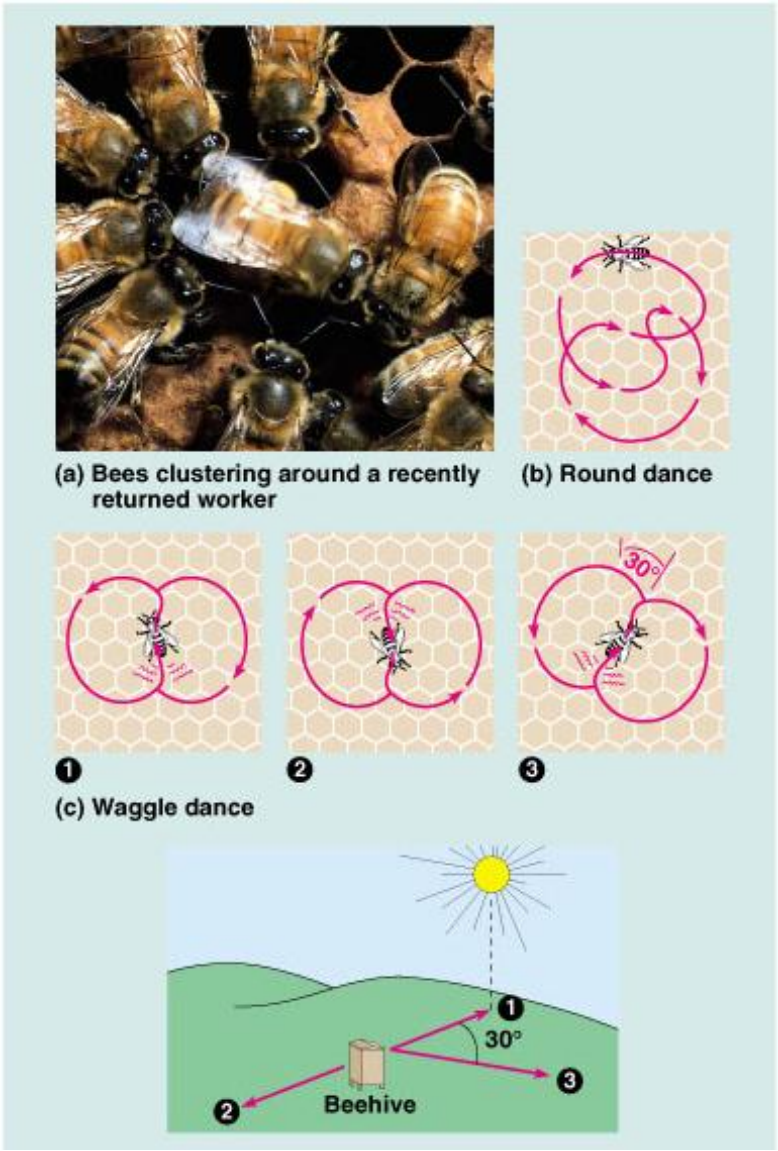
Male three-spined stickleback fish

- Male three-spined stickleback fish have red bellies
- A male perceives other males with red bellies, coming to their nesting territories as invaders
- Shows off aggressive attacking behavior

Fixed Action Patterns

- It attacks anything with red underside even unrealistic models and objects
- Red underside is a sign stimulus
- Aggressive attacking behavior is an instinct (fixed action pattern)

Fixed Action Patterns



Example

Communication in bees by Karl von Frisch

- A special dance performed by a worker bee called **“waggle dance”**
- Workers returned to the honey comb with pollen or nectar,
- Dances constitute a language that “tells” other workers where the food is.

Fixed Action Patterns

- Dance tells the direction and distance of the food source
- Worker bee uses this dance to recruit and direct other workers in gathering pollen and nectar.
- In 1973, Karl von Frisch was awarded the Nobel Prize for his research on the honey bee waggle dance.

Fixed Action Patterns



Example

Herring gull experiment by Niko Tinbergen

- Releasers (red spot on the bill) initiate behaviour automatically.
- From birth, herring gull chicks peck at a red spot on their parent's bill in order to get food.

Fixed Action Patterns

- They will also readily peck at red dots on the end of a wagging pencil, and a wagging model with the red spot on its head. Herring gull chicks respond poorly to stationary models and models with no spot.

End

Animal behaviour

Lecture # 27

Learned behaviour or
Acquired behaviour

Types of Animal behaviour

There are 2 basic types of behaviour

1. Innate behaviors or Inherent behaviour
2. Learned behaviors or acquired behaviour

1. Nature/innate

Instinct or genes determine behaviour

2. Nurture/learned

Experience and learning influence behaviour

Learned behaviour or Acquired behaviour

2. **Learned behaviors**

- modified by experience
- variable, changeable
- flexible with a complex & changing environment

Learned behaviour or Acquired behaviour

Learned (acquired) behaviour

Thorpe defined learning as a long lasting adaptive change in behaviour resulting from past experience; hence it is acquired during the life time of an individual.

Learned behaviour or Acquired behaviour

Characteristics of learned behaviour

1. Acquired during the life of an organism due to constant experience
2. Experience dependent and can be modified through experience
3. Learning is flexible
4. Learning behaviour differs from individual to individual, to individual among the same species hence not species specific

Learned behaviour or Acquired behaviour

Characteristics of learned behaviour

5. Learned behaviour is certainly not inherited, though the ability to learn is almost certainly inherited as it is dependent on the development of the nervous system of the organism which is inherited.

6. All organisms have the ability to learn at least to some extend.

7. However learning in humans surpasses all other animals. Almost everything we do has been learned. No other species surpasses the humans in their amount and range of information that is acquired through learning

Learned behaviour or Acquired behaviour

Types of learning

i. Habituation
ii. Imprinting
iii. Classical conditioning
iv. Trial and error learning
v. Latent learning
vi. Insight
vii. Reasoning
viii. Cognition

End

Animal Behavior

Lecture # 28

Habituation and Imprinting

Habituation and Imprinting

Learning Behaviors

- Behaviors that are variable with experience

Examples

- Habituation
- Imprinting

Habituation and Imprinting

Habituation

- One of the simplest forms of learning
- An animal learns to ignore a repeated, irrelevant stimulus

Definition

"A loss of responsiveness to stimuli that convey little or no new information"

Habituation and Imprinting

Habituation—Examples

- Many mammals and birds recognize alarm calls of members of their species
- If these calls prove fake repetitively, they stop responding ("cry-wolf" effect)

Habituation and Imprinting

Habituation—Examples

- A threatened squirrel gives signals
- Others recognize signal and go to nearest refuge
- An individual who gives many false alarms is also recognized
- Its signals are ignored

Habituation and Imprinting

Significance of Habituation

- An animal conserves energy and time by habituation to unimportant stimuli
- Energy and time are better spent on other important functions

Habituation and Imprinting

Imprinting

"Formation of a long-lasting behavioral attachment to a particular individual or object at a specific stage in life"

Habituation and Imprinting

Imprinting

- A type of learning behavior
- A very young animal develops attachment with the very first object it has visual, auditory or tactile experience
- It thereafter follows that object

Habituation and Imprinting

Sensitive or Critical Period

- A specific sensitive (critical) period in which imprinting occurs
- A limited developmental phase for certain learned behaviors
- Young imprint on their parent to learn basic behaviors of species
- Parent learns to recognize its offspring

Habituation and Imprinting

Imprinting—Example

- Seagulls
- Sensitive period for a parent to bond with its young lasts one to two days
- If bonding does not occur, parent will not care for the infant, leading to death of the offspring

Habituation and Imprinting

Imprinting—Example

- Classic experiment of Konard Lorenz with geese



Habituation and Imprinting

Imprinting—Example

- Konard Lorenz made goslings to imprint on him
- Goslings followed him as though he was their mother



END

Animal behaviour

Lecture # 29

Examples of Habituation and Imprinting

Examples of Habituation and Imprinting

Examples of Habituation

- Lack of continued response to strong odour is a common example of sensory habitation. Sensory system may stop after sending signals to the brain in response to a continuous repeated stimulus.

Examples of Habituation and Imprinting



Examples of Habituation

- If any disturbance is created in the spider web, the spider will response to it immediately but finding nothing it will return.
- If this stimulus is repeated several times, spider will gradually stop responding as the stimulus was of no important consequence to the spider and hence it will ignore the stimulus.
- However the spider will respond if some other end of the web is disturbed.

Examples of Habituation and Imprinting



Examples of Habituation

- Prairie dogs give alarm calls when mammals, large birds or snakes approach which warns others of the presence of these predators. These alarm calls facilitate escape into their burrows.
- However, if their burrows are located near trials used by humans, the given alarm calls every time a person walks by would be a wasted effort. Thus habituation to humans is an important adaptation to these animals.

Examples of Habituation and Imprinting



Examples of Imprinting

- When an animal learns to distinguish what an appropriate mate looks like to avoid inbreeding. It is observed in goats, zebra finches, and pandas.

Examples of Habituation and Imprinting



Examples of Imprinting

- The peregrine falcon has also been known to imprint on specific structures for their breeding grounds such as cliff sides and bridges and thus will favour that location for breeding.

Examples of Habituation and Imprinting

Examples of Imprinting

- Angelo d'Arrigo extended the technique used by Konrad Lorenz.
- He noted that the flight of hang-glider is very similar to the flight patterns of migratory birds.
- Both use updrafts of thermal currents to gain altitude that then permits soaring flight over distance.

Examples of Habituation and Imprinting

Examples of Imprinting

- He used this method to reintroduce threatened species of raptors.
- As birds hatched in captivity have no mentor birds to teach them traditional migratory routes, D'Arrigo hatched chicks under the wing of his glider and they imprinted on him. Then, he taught the fledglings to fly and to hunt.
- The young birds followed him not only on the ground (as with Lorenz) but also in the air as he took the path of various migratory routes.

Animal Behavior

Lecture # 30

Associative Learning

Associative Learning

Associative Learning

"A learning that involves making associations between experiences"

Example

- A mouse bites a brightly colored caterpillar
- It tastes bad
- Mouse avoids attacking insects of similar color in future.

Associative Learning

Associative Learning

- Ability to associate one environmental feature (such as a color) with another (such as a foul taste)

Types

- Classical conditioning
- Operant conditioning

Associative Learning

Classical conditioning

- An arbitrary stimulus becomes associated with a particular outcome

Example

- Experiments of Russian physiologist Ivan Pavlov
- He always rang a bell just before feeding a dog
- Dog conditioned to salivate at the bell's sound alone, anticipating arrival of food

Associative Learning

Operant Conditioning

"An animal learns to associate one of its own behaviors with a reward or punishment and then tends to repeat or avoid that behavior"

- also called trial-and-error learning

Associative Learning

END

Operant Conditioning: Examples

- A predator learns to avoid certain kinds of prey if associated with painful experience
- An animal may repeat an action that has resulted in a surprise finding of food
- A classic experiment: a rat was trained to obtain food by pressing a lever

Animal Behavior

Lecture # 31

Insight Learning

Insight Learning

Insight Learning

- Insight is immediate and clear learning
- understanding that takes place without overt trial-and-error testing
- Insight occurs when animals recognize relationships (or make novel associations between objects or actions) that can help them solve new problems

Insight Learning

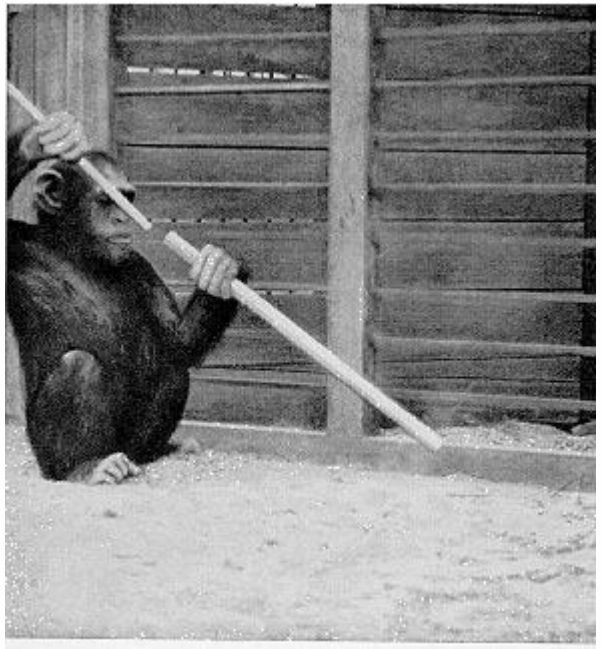
- A type of learning that uses reason, especially to form conclusions, inferences, or judgments, to solve a problem.
- It is based on advanced perceptual abilities such as thought and reasoning

Insight Learning

Kohlar's work on chimpanzees is the experimental proof of insight learning.

- During experiment Köhler placed a banana outside the cage of a hungry chimpanzee
- two sticks was given to Chimpanzee, but too short for pulling in the food but joinable to make a single stick of sufficient length

Insight Learning



- First; chimpanzee tried to use only a stick to reach banana
- Then it tried to use one stick to push the other along to touch the banana.
- Later, it accidentally joined the sticks and observed the result and used it to retrieve the banana.
- In repeated experiments, the chimpanzee joined the two sticks and took the banana immediately.

Animal Behavior

Lecture # 32

Concept of
Animal Cognition

Concept of Animal Cognition

Animal Cognition

- ability of an animal's nervous system to perceive, store, process, and use information gathered by sensory receptors
- encompasses the mental capacities of non-human animals

Concept of Animal Cognition

Animal Cognition

- Animals process information from eyes, ears, and other sensory organs to perceive the environment.
- Problem solving can be learned by observing the behavior of other animals.

Concept of Animal Cognition

Researchers have examined animal cognition in:

- mammals (primates, cetaceans, elephants, dogs, cats, horses, cattle, raccoons and rodents),
- birds (including parrots, fowl and pigeons),
- reptiles
- (lizards, snakes, and turtles)
- Fishes
- Invertebrates (cephalopods)

Concept of Animal Cognition



Examples of Animal Cognition

- Chimpanzees learn to join sticks to reach bananas
- Clark's nutcrackers can find food hidden in caches located halfway between particular landmarks.

Concept of Animal Cognition

Examples of Animal Cognition

- Ravens showing sophisticated social cognitive skills including the formation of coalitions, considering visual perspectives of others, and directing conspecifics' attention to external referents

Animal Behavior

Lecture # 33

Spatial Learning and Cognitive Maps

Spatial Learning and Cognitive Maps

Spatial Learning

“Capacity to establish a memory that reflects environment's spatial structure”

Spatial Learning and Cognitive Maps

Spatial Learning

- Spatial learning enhances an organism's fitness in environment
- Every natural environment has some spatial variation e.g.,
  - Locations of nest sites
  - Hazards
  - Food
  - Prospective mates

Spatial Learning and Cognitive Maps

Spatial Learning—Example

- Behavior of a female digger wasp
- Nests in small burrows dug into sand mounds
- A wasp covers the entrance of her nest with sand when going on hunting

Spatial Learning and Cognitive Maps

Spatial Learning—Example

- Upon return, she flies directly to her hidden nest
- Despite hundreds of other burrows in the area
- Locates her nest by memorizing its position relative to visible landmarks (location indicators)

Spatial Learning and Cognitive Maps

Spatial Learning—Example

- Proof of visual learning of object maps
- Experimentally disturbing the landmarks confuses the wasp

Spatial Learning and Cognitive Maps

Cognitive Maps

"A representation in the nervous system of the spatial relationships between objects in an animal's surroundings"

- It involves relating landmark positions to one another

Spatial Learning and Cognitive Maps

Cognitive Maps

- Cognitive maps follow abstract geometric rule—identifying halfway point between landmarks
- Reduce the amount of detail required to remember an object's location

Spatial Learning and Cognitive Maps

Cognitive Maps

- Advanced than relying solely on moving from landmark to landmark
- Using cognitive maps makes navigation more flexible and efficient

Spatial Learning and Cognitive Maps

Cognitive Map—Example

- Clark's nutcracker birds
- Store thousands of pine seeds in fall at thousands of hiding places
- Caches—distributed over a large area (35 km^2)
- During winter, they relocate their caches

END

Animal Behavior

Lecture # 34

Cognition and Problem Solving

Cognition and Problem Solving

Cognition

"The process of intellect, represented by awareness, understanding, reasoning, recalling memories and judgments"

Cognition and Problem Solving

Cognition

- Most complex form of learning
- Characteristically exhibited by primates
- Also shown by:
  - Many mammals
  - Cephalopods
  - Insects

Cognition and Problem Solving

Problem Solving

"A cognitive activity of devising a method to proceed from one state to another in the face of apparent obstacles"

- Depends on information processing ability of nervous system

Cognition and Problem Solving

Problem Solving Behavior

- Problem solving behavior is highly developed in:
- Some mammals (primates and dolphins)
- Some birds (ravens, crows, jays)

Cognition and Problem Solving

Problem Solving Behavior

- Animals may learn by observing behavior of others
- Young wild chimpanzees, copying experienced chimpanzees & learn how to crack oil palm nuts with two stones

Cognition and Problem Solving

Problem Solving: Example

- A chimpanzee placed in a room
- Several boxes on the floor
- A banana hung high out of reach
- Chimp can “size up" by stacking the boxes
- Enabling to reach food

END

Animal Behavior

Lecture # 35

Ecology and Genetics to maintain animal behaviour

Ecology and Genetics to maintain animal behaviour

Environmental Factors in Behavior Development

- Genes of an individual provides instructions for behavior development
- Expression of genes is influenced by many environmental factors
- Animal behavior develops by complex interactions between genetic and environmental factors

Ecology and Genetics to maintain animal behaviour

Environmental Factors in Behavior Development

- Environment of fertilized egg
- Animal's diet
- Animals social interactions
- Environmental conditions in animal’s surrounding

Ecology and Genetics to maintain animal behaviour

Study of Environment's Effect on Gene Expression

- Two methods to determine extent of environmental effects on expression of behavioral genes:
  - Cross-fostering study
  - Twin Studies

Ecology and Genetics to maintain animal behaviour

Cross-Fostering Study

- Young of one species are nurtured by adults of another species
- Extent of change in offspring's behavior is determined
- Changes are a measure of how social and physical environment influences behavior

Ecology and Genetics to maintain animal behaviour

Cross-Fostering Study

- Males of certain mouse species have behavioral differences that are well suited for cross-fostering studies

Ecology and Genetics to maintain animal behaviour

Twin Studies

- Influence of genetics and environment on human behavior, is explored by twin studies
- Behaviors of identical twins are compared which are:
  - Raised apart
  - Raised in the same household

Ecology and Genetics to maintain animal behaviour

Twin Studies

- Twin studies are carried out for human behavioral disorders:
  - Schizophrenia
  - Anxiety disorders
  - Alcoholism
- Reveal effect of different environments on the expression of disorder
- Vary 20-80 % in separate environments

Ecology and Genetics to maintain animal behaviour

Twin Studies

- Significant variation in development of these disorders in humans prove contribution of both environmental and genetic factors

END

Animal Behavior

Lecture # 36

Animal Signals and Communication

Animal Signals and Communication

Why study animal communication?

- Central to the study of animal behavior
- Tool for elucidating general evolutionary principles

– Taxonomy, reproductive isolation, speciation, natural selection

Animal Signals and Communication

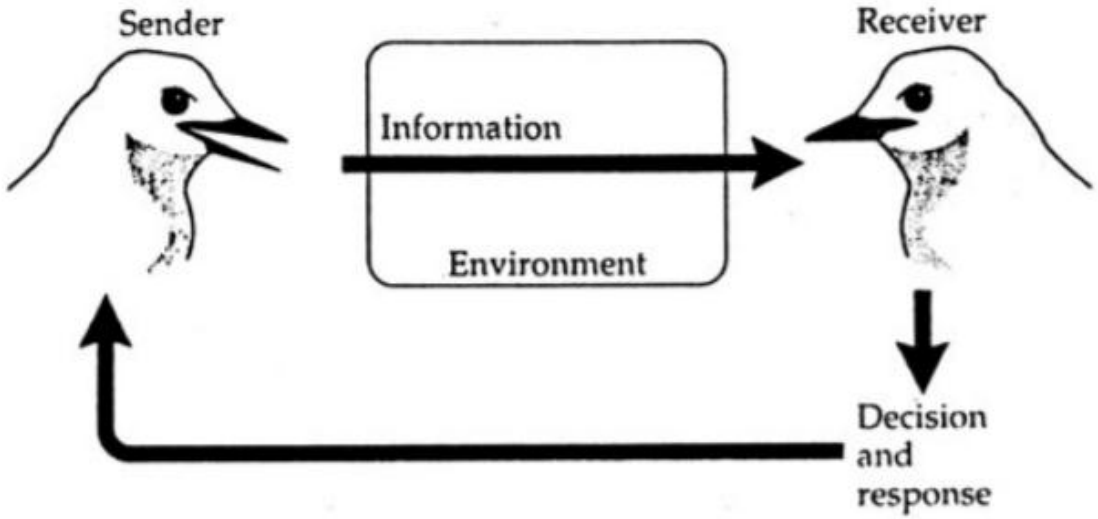
Why study animal communication?

- Has practical and conservation importance
  - – Censusing populations and communities
  - – Insect pest control via mate attraction signals
  - – Welfare of captive and wild animals

Animal Signals and Communication

- There is a sender and a receiver in communication.
- Study of animal communication is called **Zoosemiotics.**
- Classical ethological view was developed by **Niko Tinbergen.**

Animal Signals and Communication



Communication

"Communication is the transfer of information via signals sent in a channel between **sender** and a **receiver** causing some kind of change in the animal that gets the information.

Animal Signals and Communication

2 Types of Communication

- **Intraspecific Communication**

Communication within a single species, e.g., honeybee dancing

- **Interspecific Communication**

Communication among different species

Prey to predator e.g., warning colouration in wasps

Predator to prey e.g., Angler fish

Human/animal communication during domestication

Conti…

Animal Behavior

Lecture # 37

Animal Signals and Communication

Animal Signals and Communication

Animal Signals

"A stimulus transmitted from one animal to another"

- Signals may be:
  - Visual
  - Chemical
  - Tactile
  - Auditory

Animal Signals and Communication

Animal Communication

- Transmission and reception of signals
- Essential element of interactions between individuals
- Used to transmit information among individuals

Animal Signals and Communication

Communication Patterns

Correlated with signals:

- Visual communication
- Olfactory communication
- Tactile communication
- Auditory communication

Animal Signals and Communication

Using Form of Communication

- Closely related to an animal's lifestyle and environment

Animal Signals and Communication

Examples—Form of Communication

- Diurnal animals communicate primarily by visual and auditory signals
- In a nocturnal terrestrial mammals visual displays are relatively ineffective
- They use olfactory and auditory signals—work well in dark

Animal Signals and Communication

Example of Animal Communication

- Courtship behavior of fruit fly *Drosophila melanogaster*
- Constitutes a stimulus-response chain
- Response to each stimulus is itself stimulus for the next behavior

Animal Signals and Communication

Courtship Behavior of Drosophila

- A male identifies a female of its species—visual communication
- Male detects chemicals released by female by smell–olfactory or chemical communication
- Male approaches and touches the female with a foreleg—tactile communication

Animal Signals and Communication

Courtship Behavior of Drosophila

- Male extends and vibrates wings: courtship song—auditory communication

Result of Successful Communication

- Female identifies male as member of the same species
- Becomes receptive
- Allows male to attempt copulation

Animal Signals and Communication

Communication for Information Transfer

- Information content varies considerably

Example

- Symbolic language of honeybees
- Share information about the location of food sources

Animal Signals and Communication

Symbolic Language of Honeybees

- A honeybee finds a food source close to the hive (less than 50 m away)
- Returns to its hive
- Moves in tight circles, moving its abdomen from side to side "round dance”
- Motivates other bees to leave the hive
- Follow her for nearby food

Animal Signals and Communication

END

Symbolic Language of Honeybees

- When food source is farther
- “Waggle dance” is performed
- Swings half-circle in one direction, a straight run, a half-circle swing in other direction
- Communicates direction and distance of source

Animal Behavior

Lecture # 38

Pheromones and Chemical Communication 1

Pheromones and Chemical Communication

Pheromones

- Chemical messengers released by many animals in environment to communicate
- Comprise 2 Greek words: “Pherein”(to transfer) and “Hormon” (to excite)

Pheromones and Chemical Communication

Pheromones

- Spread through air
- Affect the behavior of individuals of a species
- Effective at remarkably low concentrations
- Have particular odors
- Detected by olfactory receptors of receiving animal

Pheromones and Chemical Communication

Chemical Communication

- Chemical communication through pheromones is well developed in:
  - Unicellular organisms
  - Insects
  - Fishes
  - Salamanders
  - Mammals

Pheromones and Chemical Communication



Examples

- **Queen bee** gives off a pheromone that prevents other females in the hive from becoming sexually mature, with the result that only the queen bee mates and lays eggs.

- Bees release an alarm hormone right after stinging a predator.

Pheromones and Chemical Communication



Examples

The **mountain lion** is a solitary animal and it likes to travel alone. A male's territory can be more that 100 square miles. The male mountain lion often kicks up piles of pine needles, dirt or leaves with its hind feet and urinates on the pile to mark its territory.

Pheromones and Chemical Communication



Examples

The **Canada lynx** also leaves a scent post. It will urinate on trees and tree trunks. It may also leave claw marks on trees. Claw marks are a visual sign, but can also be a chemical one. Many animals have scent glands in their claws. When they scratch a tree, they leave a little of their scent behind.

Pheromones and Chemical Communication



Examples

Some animals use scent to identify members of their family. The **mule deer** has scent glands on its hind legs right above its hooves. Young deer will often identify their mothers by sniffing at the glands. When mule deer are in groups, they often sniff each others hind legs. When male mule deer become aggressive towards each other, the hair around the glands will stand up.

Pheromones and Chemical Communication



Examples

Domestic cats have scent glands near their mouths, on their foreheads and at the base of their tails. They use these glands to mark territory.

Pheromones and Chemical Communication

Examples

Many animals use chemical communication or pheromones to communicate. They may leave their scent to mark territory or attract mates. They often use smell to find prey and identify other animals.

Pheromones and Chemical Communication

Examples

Some animals leave **scent posts** to mark their territory. A scent post is something that is marked by an animal with its scent. Scent posts set the boundaries of an animal's territory.

Pheromones and Chemical Communication



Examples

- Some animals leave **scent posts** to mark their territory. A scent post is something that is marked by an animal with its scent. Scent posts set the boundaries of an animal's territory.

- Ants use trail forming or recruitment pheromones that are useful in foraging.

Animal Behavior

Lecture # 39

Pheromones and Chemical Communication 2

Pheromones and Chemical Communication

Roles of Pheromones

1. Recognize Members of Species

- Unicellular organisms recognize members of their own species
- Many insects identify members of their colony

Pheromones and Chemical Communication

Roles of Pheromones

2. Reproductive Roles

- Pheromones attract members of opposite sex for breeding
- Induce courtship behaviors in members of a species
- Induce spawning behavior in aquatic animals with external fertilization

Pheromones and Chemical Communication

Roles of Pheromones

3. Territorial Behaviors

- Many male mammals mark their territories with pheromones
- It warns other males of their occupied area

4. Insect's Social Order

- In a honeybee colony, pheromones produced by queen maintain hive's complex social order

Pheromones and Chemical Communication

Roles of Pheromones

5. Alarm Calls

- Members of some species produce pheromones on getting injured
- It warns other members of presence of danger in area

Pheromones and Chemical Communication

Roles of Pheromones

6. Repel Predators

- Pheromones may repel predators
- Example: foul-smelling musk that makes skunks unpalatable to their enemies

Pheromones and Chemical Communication

Advantages of Pheromones as Chemical Signal

- Provide a simple message that can last for hours or days
- Are equally effective during night or day
- May be transported over long distances by air
- Take relatively little energy to produce

Pheromones and Chemical Communication

Disadvantages of Chemical Signals

- Chemical signals are slow to act
- They cannot be changed quickly

END

ZOO 403
Animal Behavior

Final Term
Lecture # 40 to 72

Compiled by: **Atiq Rehman**

Animal Behavior

Lecture # 40

Behavioral Rhythms

Behavioral Rhythms

Biological Clocks

Animals respond rhythmically to cyclic fluctuations in environment e.g.

- Changes in seasons
- Daily light and dark periods

Rhythmic behaviors are controlled by biological clocks

Behavioral Rhythms

Biological Clocks

- Biological clocks regulate daily and seasonal cyclic activities
- Clocks have a molecular genetic basis
- Genes for time sense have been identified in many animals (human beings too)

Behavioral Rhythms

Biological Clocks

Biological clocks may be:

- Circadian clocks
- Circannual clocks
- Lunar clocks

These clocks are influenced by:

- Dark and light periods
- Lunar cycles

Behavioral Rhythms

Circadian Clock and Circadian Rhythms

- Most important biological clock in all animals
- Regulates and coordinates daily behavioral activities
- Its output is a circadian rhythm
- Exhibited as a daily cycle of rest and activity

Behavioral Rhythms

Circadian Clock and Circadian Rhythms

- Clock is synchronized with light and dark cycles of the environment
- Can also maintain rhythmic activity under constant environment conditions (e.g. during hibernation)

Behavioral Rhythms

Circannual Rhythms

- Biological rhythms linked to yearly cycle of seasons
- Span events repeating on 12 month basis
- Correspond to tilt of earth's spin axis resulting in seasonal changes
- Circannual behaviors:
  - Hibernation
  - Migration
  - Reproduction

Behavioral Rhythms

Lunar Cycles and Tidal Rhythms

- Controlled by phases of the moon
- Gravitation pull of moon determines timing of tidal rhythms in marine environments

Behavioral Rhythms

Lunar Cycles and Tidal Rhythms

- Reproductive behaviors of many marine animals are linked to the timing of new or full moon
- This behavior links reproduction to the times of greatest tidal movement

Behavioral Rhythms

Lunar Cycles and Tidal Rhythms

- Tides disperse larvae to deeper waters
- Deeper waters are relatively safer environment to complete early development

END

Animal Behavior

Lecture # 41

Maintenance of Internal Biological Clocks

Maintenance of Internal Biological Clocks

What are biological clocks?

- Biological clocks are organisms' natural timing devices, regulating the cycle of circadian rhythms.
- They're composed of specific proteins that interact with cells throughout the body.

Maintenance of Internal Biological Clocks

What are biological clocks?

- Nearly every tissue and organ contains biological clocks.
- Researchers have identified similar genes in people, fruit flies, mice, plants, fungi, and several other organisms that make the clocks' molecular components.

Maintenance of Internal Biological Clocks

Biological Clocks

- In humans (and other mammals), a circadian clock is located in the suprachiasmatic nuclei (SCN).
- The SCN is in the hypothalamus. It is a tiny cluster of about 10 thousand nerve cells.

Behavioral Rhythms

Biological Clocks

- The internal mechanism by which such a rhythmic phenomenon occurs and is maintained even in the absence of the apparent environmental stimulus is termed a **biological clock.**
- This biological clock maintain the rhythmic cycle in all animals e.g., mammals and birds.

Maintenance of Internal Biological Clocks

Biological Clocks

- When an animal that functions according to such a clock is rapidly translocated to a geographic point where the environmental cycle is no longer synchronous with the animal's cycle, the clock continues for a time to function synchronously with the original environmental cycle.

Maintenance of Internal Biological Clocks

Mechanism and maintenance

- This group of cells in the hypothalamus respond to light and dark signals.
- When our eyes perceive light, our retinas send a signal to our SCN.
- The SCN sets off a chain reaction of hormone production and suppression that affects body temperature, appetite, sleep drive, and more.

Maintenance of Internal Biological Clocks

Mechanism and maintenance

- Each morning, as sunlight creeps in, our body temperature begins to rise and cortisol is released, increasing our alertness and causing us to wake up.

Maintenance of Internal Biological Clocks

Mechanism and maintenance

- In the evening, as it becomes dark outside, melatonin levels rise and body temperature lowers.
- Melatonin stays elevated throughout the night, promoting sleep.

Maintenance of Internal Biological Clocks

Mechanism and maintenance

- As long as our eyes perceive light, the SCN responds by suppressing melatonin production.
- This explains why evening exposure to light, such as that from indoor light or electronic devices that emit blue light, such as a computer or television, make it harder to fall asleep.

END

Animal Behavior

Lecture # 42

Animal Navigation and Migration

Animal Navigation and Migration

Migration

- A regular, long-distance change in location
- Involves navigation over long distances
- Travelling through unfamiliar territories and environments—not previously encountered

Animal Navigation and Migration

Stimuli for Migration

Environmental and seasonal stimuli:

- Seasonal droughts
- Harsh weather conditions
- Hot summers
- Cold winters

Animal Navigation and Migration

Migratory Animals

Migratory behavior observed in a wide variety of:

- Birds
- Fishes
- Butterflies
- Grey whales
- Many other animals

Animal Navigation and Migration

Migratory and Navigational Cues

- Navigational abilities rely on several sensory cues (signals)
- Guide in orientation and path finding
- Some types of cues are not perceivable by human beings

Animal Navigation and Migration

Migratory and Navigational Cues

- Sun compass
- Celestial cues
- Earth's magnetic field
- Visual cues
- Auditory cues
- Olfactory cues

Animal Navigation and Migration

Sun Compass

- Tracking position relative to the sun
- Sun's relative position changes during the day
- Animals adjust for these changes through circadian clock
- An internal mechanism that maintains a 24-hour activity rhythm or cycle

Animal Navigation and Migration

Celestial Cues

- Nocturnal navigators
- North Star in the night sky—has a constant position
- Navigators use this cue

Animal Navigation and Migration

Earth's Magnetic Field

- Long-distance migrants have magnetic sense
- Navigate by sensing position relative to earth's magnetic field
- e.g. birds and fishes

Animal Navigation and Migration

Visual, Auditory and Olfactory Cues

To find way to their destination, many birds use:

- Particular landmarks
- Plane of polarized light
- Odors
- Sounds

END

Animal Behavior

Lecture # 43

43. Competition for Resources

Competition for Resources

How do organisms compete for resources?

- **Competition** is the struggle among organisms to survive in a habitat with limited resources (species searching for the same kinds of food).

- Competition happens when organisms in an ecosystem have similar needs.

Competition for Resources

How do organisms compete for resources?

- Organisms with different needs can live and survive together in a habitat with little competition.
- Competition increases when the amount of available resources is low.
- Members of the same species compete for available resources.
- All organisms compete for resources (not just animals).

Competition for Resources

Competition occurs

- Within species (intraspecific competition)
  - ✓ Can limit population size
  - ✓ Can affect patterns of spatial dispersion

Competition for Resources

Competition occurs

- Between species (interspecific competition)
  - ✓ Can limit population size
  - ✓ Can affect patterns of spatial dispersion
  - ✓ Can influence patterns of diversity
  - ✓ Can act as a selective force on traits

Competition for Resources

What do organism compete for ?

- Organisms compete for the resources they need to survive
  - ✓ Air
  - ✓ Water
  - ✓ Food
  - ✓ space

Competition for Resources

What do organism compete for ?

- Animals and plants that have specific life history requirements, like cavity-nesting birds, or animals with obligate feeding behaviors, have a more difficult time competing.
- These resources can be limiting factors for where organisms are distributed, and competition for them can be fierce.

Animal Behavior

Lecture # 44

44. Habitat Selection

Habitat Selection

Habitat

- A habitat is a place where an organism makes its home.
- A habitat meets all the environmental conditions an organism needs to survive.
- For an animal, that means everything it needs to find and gather food, select a mate, and successfully reproduce.

Habitat Selection

Habitat selection

- To live in a habitat an animal must first have **access to the habitat**.
- Once the animal has access to the habitat it must be able to tolerate the conditions of the habitat and find the resources that it needs to survive in that habitat.

Habitat Selection

Habitat selection

- Animals must **be able to tolerate** at least two kinds of factors in the habitat. These factors are abiotic factors and biotic factors.
  - ✓ Abiotic factors are non-biological factors such as temperature, humidity, salinity and pH to name a few.
  - ✓ Biotic factors are biological factors such as competition, predation, and disease.

Habitat Selection

Habitat selection

- Then the animal has **to find the resources** that it needs to survive.
- These recourses include: shelter, food, protection, mates, space for breeding, feeding, resting, roosting, courtship, grooming, sleeping etc.

Habitat Selection

Habitat selection

- Animals have **to compete for recourses** too. Two species or more occur in a single habitat, in which they strive for survival, shelter, food; space etc. there may be considerable competition, predation, diseases, allelopathic agents (antibiosis/ poisoning) in the existing habitat.

- Every available space cannot play a role of habitat for all species at a time due to certain antagonistic factors. Therefore the animals select the habitat which fulfils its niche and keeps it away from other antagonistic factors.

Habitat Selection

Birds are ideal for studies of habitat selection as they are highly mobile, often migrating thousands of miles (and in the process passing over an enormous range of environments), and yet ordinarily forage, breed, and winter in very specific habitats. For Example

- ✓ The small migrant songbirds
- ✓ Male Henslow's Sparrows foraging
- ✓ Female Red-eyed Vireos seeking their food

Animal Behavior

Lecture # 45

Foraging Behavior

Foraging Behavior

Foraging Behavior

- All activities of an animal related to search, recognize and capture food items

Foraging Behavior

Foraging Choices for Animals

- What items should be included in the diet?
- What path should an animal take between patches?
- How should it locate new patches of food?
- If food is depleted in a patch, when should animal depart to seek another patch of food?

Foraging Behavior

Foraging Choices for Animals

- All animals require decisions among such choices—e.g.,
- Hummingbirds and bees, visiting clumps of flowers to obtain nectar
- Owls, foraging for small rodents in different habitats (fields and forests)

Foraging Behavior

Evolution of Foraging Behavior

- Nutrition is essential for an animal's survival and reproductive success
- Natural selection has operated to refine behaviors that enhance efficiency of feeding

Foraging Behavior

Foraging and Role of Genetic Components

- Variation in forager (*for*) gene in Drosophila dictates food search behavior in larvae
- Larvae carrying allele for^R (Rover) travel twice for feeding than with for^s (sitter)

Foraging Behavior

Foraging and Role of Genetic Components

- Enzyme encoded by forager locus is involved in signal transduction pathway
- It is more active in *for*^{R} than in *for*^{S} larvae
- Result: genetic components are behind foraging behaviors

END

Lecture # 46

Foraging Model: Risk and Reward Balance

Foraging Model: Risk and Reward Balance

Gains and Costs of Foraging

- Gains and costs in finding and consuming food:
  - Energy gain from food
  - Energy expenditure in foraging
  - Risk of predation while foraging

Foraging Model: Risk and Reward Balance

Optimal Foraging Model

- Behavioral ecologists apply a model of cost-benefit analysis to study foraging strategies of animals
- Animal foraging behaviors reflect a compromise between competing selective pressures i.e.
  - Benefits of nutrition
  - Costs of obtaining food

Foraging Model: Risk and Reward Balance

Optimal Energy Expenditure

- Optimal Foraging Model analyzes:
  - Energy needed to search food
  - Energy needed to pursue and handle food
  - Energy required to digest food

Foraging Model: Risk and Reward Balance

Optimal Energy Expenditure

- For an animal, energy gain from a given food item must exceed energy costs
- According to optimal foraging model, natural selection favors a foraging behavior that:
  - Minimizes costs of foraging
  - Maximizes benefits

Foraging Model: Risk and Reward Balance

END

Balancing Risk and Reward

- Most potential cost to a forager is risk of predation
- Maximizing energy gain and minimizing energy costs have no importance if the forager becomes meal of a predator
- Animal's foraging behaviors reflect more safety from predation risk and not the hunger for more food gain

Animal Behavior

Lecture # 47

47. Specialists and Generalists

Specialists and Generalists

Species Diversity

- Species can be broadly divided into:
  - ✓ Generalist species
  - ✓ Specialist species

Specialists and Generalists

Generalist species

- Species that can live in many different types of environments, and have a varied diet are considered **generalists**.

Specialists and Generalists

Generalist species

- Generalists can eat a variety of foods
- Thrive in a range of habitats.
- They have broad niches.
- They can thrive in rapidly changing environmental conditions.

Specialists and Generalists

Examples of Generalist species

- **Raccoons** (*Procyon lotor*) are an example of a generalist species. They can live in a wide variety of environments, including forests, mountains, and large cities, which they do throughout North America. Raccoons are omnivores and can feast on everything from fruit and nuts to insects, frogs, eggs, and human trash.

- Other examples of generalist species include bobcats, coyotes, cockroaches, rats, mice, flies, white-tailed deer and humans, etc.

Specialists and Generalists

Specialist species

- **Specialist** species are animals that require very unique resources. Often, these species have a very limited diet, or need a specific habitat condition to survive.

Specialists and Generalists

Specialist species

- Specialist species evolved to fit a very specific niche.
- Environmental disruptions like effects from climate change or habitat loss have a strong effect on specialists.
- They cannot adapt to use other food sources or habitats as quickly as generalist species.
- The number of specialist species is declining due to human activities.

Specialists and Generalists

Examples of Specialist species

- **Tiger salamanders** can only reproduce in wetland habitats. They also require an abundance of insects and worms for their diet.

- **Koala** (*Phascolarctos cinereus*) are native to Australia. They feed only on the leaves of the eucalyptus tree. Therefore, their range is restricted to habitats that support eucalyptus trees.

- **Canada lynx** (*Lynx canadensis*) a carnivore only feed on snowshoe hare. It inhabit the forested and mountainous areas where their prey is present.

Specialists and Generalists

	Generalist	Specialist
DEFINITION	Generalist is a species that eats a variety of food and lives in a wide range of environmental conditions	Specialist is a species that has a limited diet and lives only in a narrow range of environmental conditions
ENVIRONMENTAL CONDITIONS	Thrives in a wide variety of environmental conditions	Thrives only in a narrow range of environmental conditions
DIET	Eats a variety of food	Has a limited diet
HABITAT	A range of habitats	Strict to a habitat
VARYING ENVIRONMENTAL CONDITIONS	Can adapt to varying environmental conditions	Unable to adapt to varying environmental conditions
EXAMPLE SPECIES	Raccoons, bobcats and coyotes, etc.	Koala, Canada lynx, Cactus and pandas
NUMBER OF GENERALIST SPECIES	Increases	Declines

Lecture # 48

Mating Behaviors and Systems

Mating Behaviors and Systems

Mating Behavior

- Mating behaviors enhance reproductive success of a species
- Mating behaviors of animal species evolved through sexual selection—a form of natural selection

Mating Behaviors and Systems

Mate Choice

- Mating behavior involves mate choice
- Mate choice includes:
  - Seeking or attracting mates
  - Choosing among potential mates
  - Competing for mates

Mating Behaviors and Systems

Mating Systems

- Mating systems of different species vary
- They are of three types:
  - Monogamous mating system
  - Polygamous mating system
  - Promiscuous mating system

Mating Behaviors and Systems

Monogamous Mating Systems

- In a monogamous relationship:
  - One male mates with one female during a breeding season
  - Mates usually remain together for a longer period

Mating Behaviors and Systems

Monogamous Mating Systems

- Monogamy is rare in most animal groups
- It is typically found in groups exhibiting nesting behaviors and parental care
- It is most frequent In birds
- More than 90% birds are monogamous

Mating Behaviors and Systems

Monogamous Mating Systems

- In a few bird species, partners are chosen for life and often remain together throughout the year (e.g. swans and geese)
- In others, seasonal monogamy is more common

Mating Behaviors and Systems

Polygamous Mating Systems

- An individual of one sex mates with several of the other
- Polygamous relationships are of two types:
  - Polygynous Systems
  - Polyandrous Systems

Mating Behaviors and Systems

Polygynous Systems

- Involve an exclusive relationship of one male with two or more females
- Polygynous species are generally dimorphic
- Males are often showier and larger than females

Mating Behaviors and Systems

Polyandrous Systems

- One female has an exclusive relationship with two or more males
- Polyandrous species are also dimorphic
- Females are generally more ornamented and larger than males

Mating Behaviors and Systems

Promiscuous Mating System

- A member of one sex within the social group mates with any member of the opposite sex
- There are no strong pair bonds or lasting relationships

Mating Behaviors and Systems

Promiscuous Mating System

- In promiscuous species, males and females are often morphologically alike
- They may be difficult or impossible to distinguish based on external characteristics

END

Lecture # 49

Sexual Selection and Mate Choice

Sexual Selection and Mate Choice

Sexual Selection

- Sexual selection among members of a species may be:
  - Intersexual selection
  - Intrasexual selection

Sexual Selection and Mate Choice

Intersexual Selection

- Involves mate choice
- Members of one sex choose mates with particular characteristics, e.g.
  - Appearance
  - Color
  - Courtship songs

Sexual Selection and Mate Choice

Intersexual Selection

- Intersexual selection has led to sexual dimorphism in secondary sexual characteristics, e.g.
  - Ornate plumage of birds
  - Antlers of deer
  - Manes of lions
- Selective characters correlate in general with the mate's health and vitality

Sexual Selection and Mate Choice

Intersexual Selection

- Purpose of mate choice is to choose a healthy partner likely to enhance reproductive success
- Mate choice patterns vary among species
- In some, females make choice for their mates
- In others, males make selections among the available pool of females

Sexual Selection and Mate Choice

Intrasexual Selection

- Involves competition for mates b/w members of same sex
- To access to members of opposite sex
- Usually occurs b/w males
- May involve male-to-male combat or fight
- Such individuals have developed weapons for competition (e.g. horns, antlers etc.)

Sexual Selection and Mate Choice

Intrasexual Selection

- Two main types of competition among males over females:
  - Scramble
  - Contest competition
- Scramble: race to get to female first and grab her
- Male body size and vigor contribute to success

Sexual Selection and Mate Choice

Intrasexual Selection

- Contest is a more typical form of competition
- Male with best fighting technique, largest body size or largest weapons will win the female

END

Lecture # 50

50. Dominance Hierarchies

Dominance Hierarchies

Dominance Hierarchy

- A form of animal social structure in which a linear or nearly linear ranking exists, with each animal dominant over those below it and submissive to those above it in the hierarchy.
- Dominance hierarchies are best known in social mammals, such as baboons and wolves, and in birds, notably chickens.

Dominance Hierarchies

Dominance Hierarchy

- Dominance hierarchies are a common among organisms.
- Some individuals will behave more dominantly toward others, these interactions result in a rank or status order.
- In some species, social ranks do not change (e.g. in insects such as bees), but most social mammals like humans and other primates have adaptable roles, where one individual can move through many social ranks over a lifetime.

Dominance Hierarchies

Dominance Hierarchy

- The study of animal sociality is called **sociobiology**, which E. O. Wilson (1975) defines as “the extension of population biology and evolutionary theory to social organization”.
- Dominance hierarchies impact the survival and reproduction of individuals by regulating access to food, mates, and shelter.
- As such, the stability of a dominance hierarchy can affect whether a population prospers or collapses.

Dominance Hierarchies

Dominance Hierarchy

- Dominance hierarchy is relatively stable from day to day.
- Direct conflict is rare; an animal usually steps aside when confronted by one of higher rank.

Dominance Hierarchies

Dominance Hierarchy

- Temporary shifts occur; for instance, a female baboon mated to a high-ranking male assumes a high rank for the duration of the pair bond.
- An individual weakened by injury, disease, or senility usually moves downward in rank.

End

Animal Behavior

Lecture # 51

51. Examples of Dominance Hierarchies

Examples of Dominance Hierarchies

Dominance Hierarchies

- Dominance hierarchies involve a ranking of individuals in a group (a “pecking order”).
- Alpha, beta rankings exist.
- The alpha organisms control the behavior of others.

Examples of Dominance Hierarchies

Examples:

1. Dominance Hierarchy in Monkeys

- The troop always has a dominant male (**alpha male**).
- The **crowning** of alpha male occurs with threats and attack.
- The **threat posture** is seen in a typical male is an open mouth with teeth exposed, ears are held against head.

Examples of Dominance Hierarchies

Dominance Hierarchy in Monkeys

- The head and body is held stiff.
- The attack includes biting and manual biting.
- There will not be much injury caused to other.

Examples of Dominance Hierarchies

Dominance Hierarchy in Monkeys

- Staring is also a part of competition.
- The male stares at another male with head held forward and brow protruded.
- The loser walks away.

Examples of Dominance Hierarchies



Dominant male



Subordinate male

Dominance Hierarchy in Monkeys

- The dominant male moves around majestically with its tail raised in the form of S letter.
- The subordinate male assumes a crouching or subdued position.

Examples of Dominance Hierarchies

Examples:

2. Dominance Hierarchy in wolf packs

- Dominance is highly organized in these species
- These animals hunt in groups and are keenly aware of their own status in relation to the other individuals present within the group.

Examples of Dominance Hierarchies

Examples:

- Dominant wolves are referred to as **alphas**.
- There are commonly both alpha males and females.
- The alpha status assures a wolf the highest priority in getting every resource e.g. food.
- These factors determine the survival of a particular individual and their ability to pass on their genes.

Examples of Dominance Hierarchies



Examples:

3. Dominance Hierarchy in Lion prides

- In lion prides, an alpha male or sometimes multiple alpha males (that are closely genetically related) hold dominant positions in the pride.

Examples of Dominance Hierarchies

Dominance Hierarchy in Lion prides

- An animal in the alpha position spends much of his time in scent marking the territory and patrolling its boundaries in order to prevent rival males from infringing upon the territory.
- The alpha male will be have priority over the choice of food and mating, thereby enhancing his overall fitness.

Examples of Dominance Hierarchies

Dominance Hierarchy in Lion prides

- Female lions also vary in dominance status.
- Dominance can be determined in part by age and hunting prowess.
- The alpha status regularly changes over when an animal becomes sick, older, or simply when a stronger animal moves in and successfully challenges the leader.

End

Animal Behavior

Lecture # 52

Agonistic Behaviors

Agonistic Behaviors

Agonistic Behaviors

- Behaviors involving confrontation, aggression, threat displays, attacks and fights between animals
- Displayed by individuals of many animal species
- Reason: limited resources e.g. food, shelter and mates

Agonistic Behaviors

Agonistic Behavior

- These behaviors are usually used only to intimidate the enemies
- They are rarely lethal
- These seemingly antisocial behaviors are important in maintaining:
  - Social order
  - Territories
  - Dominance hierarchies

Agonistic Behaviors

Two Types of Agonistic Behaviors

- Territorial Behaviors
- Dominance hierarchies

Territorial Behaviors

- Include territorial ownership and defense
- A territory is a fixed area in which an animal spends most of its time and defends it from intruders of the same species

Agonistic Behaviors

Territorial Behaviors

- Found in most social animal populations e.g.
  - Insects
  - Crustaceans
  - Fishes
  - Amphibians
  - Lizards
  - Birds
  - Mammals, including humans

Agonistic Behaviors

Purposes of Territorial Behaviors

- Occupying a defined territory for breeding (as by many male birds and mammals)
- Male actively defends his area against other males
- He attracts a female in his territory and performs courtship and mating without interference

Agonistic Behaviors

Purposes of Territorial Behaviors

- Occupying a territory with abundant food supply
- Occupying a territory that provides shelter to the animal from predators and unfavorable climate

Agonistic Behaviors

Dominance Hierarchies

- "Organization of groups in such a way that some members of the group have greater access to resources (food, mates) than others"

Agonistic Behaviors

Dominance Hierarchies

- Some animals in a group are considered at top of the order
- They have first choice of resources
- Members present near the bottom do not get sufficient resources
- Such hierarchies exist in many vertebrate groups

END

Lecture # 53

53. Examples of Agonistic Behaviors

Agonistic Behaviors

Agonistic Behaviors

- Any social behaviour related to fighting.
- It includes threats, displays, retreats, placation, and conciliation.

Examples of Agonistic Behaviors



Examples

1. Western gorilla (*Gorilla gorilla*)

- Male displays a wide range of both vocal and gestural communications when threatened by an opponent.
- A silverback (alpha male) will start hooting, throwing, chest pounding, leg kicks, and sideways running when approached by another male.
- This is done to intimidate the opponent and show physical abilities without actually making any physical contact.

Examples of Agonistic Behaviors



Examples

2. Sheep

- Often viewed as non-aggressive, gregarious animals but they do show agonistic behaviour.
- Aggressive behaviour include shoving with shoulders, running together and butting.
- Play butting occurs in young lambs and more damaging aggression in ewes and rams.

Examples of Agonistic Behaviors

Examples

- Most fighting is during breeding season.
- Threat postures include: striking the ground, tooth grinding, lateral body presentation, sniffing, mounting and chasing.
- Aberrant aggression: Some ewes show aberrant aggression towards their lambs or alien lambs.
- Wool picking with teeth is common in confined sheep.

Examples of Agonistic Behaviors



3. Chickens

- Chickens show threats associated with fighting, leaping and wing-flapping.
- The major aggressive act is pecking.
- Submissive behaviours are described as retreat.
- Aberrant behaviour can be found among confined chickens.
- Cannibalism could be influenced by diet and management procedures.

Examples of Agonistic Behaviors

4. Cattle

- Pre-fight behaviours – Active or passive avoidance leads to fight.
- Close contact, head lowered, ready to fight and butting or active fighting.
- Butting could be regarded as a non-retaliated blow with the head, while fighting involves reciprocal butts, circling and pushing.

Examples of Agonistic Behaviors

- The end of the fight begins with one animal showing submissive behaviour.
- Aberrant agonistic behaviour include naval sucking, fence and pen chewing (aggression towards inanimate objects) and ear sucking.

Examples of Agonistic Behaviors



5. Black mamba (*Dendroaspis polylepis*)

- Agonistic fighting for black mambas involves a wrestling match.
- Opponents attempt to pin each other's head repeatedly to the ground.
- Fights normally last a few minutes but can extend to over an hour.
- The purpose of fighting is to secure mating rights to receptive females nearby during the breeding season.

Examples of Agonistic Behaviors





End

6. Frill-necked lizard

- It uses its frill as a way to display size and aggression.
- In comparison to its body size, frill can flare out to make the lizards head look several times bigger.
- It displays bright orange and red scales.
- Males fight and display frills often during the mating seasons.
- Male display includes repeated partial erections of frill, head bobbing, tail lashing, and waving of forelimbs.

Animal Behavior

Lecture # 54

54. Social Organization

Social Organization

Social Organization

- A **society** is a group of individuals belonging to the same species and organized in a cooperative manner.
- The members of a social organization are:
  - ✓ interdependent
  - ✓ variously specialized in function and
  - ✓ survival depends on co operation in the group.

Social Organization

Social Organization

- A social organization is a large group of members who:
  - intercommunicate extensively
  - exhibit division of labour
  - bear some relatively permanent social relationship to one another
- Such organization exists in both invertebrates and vertebrates.

Social Organization

Examples

- Among vertebrates, arthropods such as ants, bees and termites have well organized colonies.

- A well established social organization is seen in primates such as monkeys, baboons, langurs etc.

Social Organization

Origin of Social Organization

- The social organization could have been originated due to the following reasons:

    1. The offspring chose to stay with their parents for protection, obtain food, learning, communication, survival, etc.

    2. When two adults choose to stay together. It can be for survival, sharing of work, mating, protection from predators, social behaviours like altruism and kin selection

Social Organization

Size and composition of a social organization

We can understand the organization size and composition with the help of an example of Bonnet Monkey.

- The Bonnet monkey, *Macaqua radiate* is the most common macaque of south India.
- It can be easily recognized by a cap of brown hair that appears like a baby's bonnet (cap).

Social Organization

Size and composition of a social organization

- There is sexual dimorphism and males are larger than female with distinct canines.
- They live in groups called **troops** and live in forests, villages and also in cities.
- A troop consist of 30 individuals with both males and females of different age groups.

Social Organization

It includes:

a. **Infants:** They have chocolate colour hair

b. **Juveniles:** between 1-2 years with brown hair

c. **Seniors:** between 2-3 years with brown hair

d. **Sub adults:** 3-5 years old males and 2-3 years old females

e. **Adults.** 4 years old males and 3 years old females. Have brown hair on the back and white hair on belly. The older individuals have yellowish grey hair.

End

Lecture # 55

55. Benefits of Social Organization

Benefits of Social Organization

Benefits of Social Organization

There are many benefits of living in social organization.

Few of them are discussed here:

1. One obvious benefit for social aggregations is **defense, both passive and active, from predators.**

Examples:

- Musk-oxen that form a passive defensive circle when threatened by a wolf pack are much less vulnerable than an individual facing the wolves alone.

Benefits of Social Organization

Examples:

- A breeding colony of gulls, alerted by the alarm calls of a few, attack predators *in group;* this collective attack is certain to discourage a predator more effectively than individual attacks.
- Members of prairie dogs, although divided into social units called **coteries**, cooperate by warning each other with a special bark when danger threatens. Thus they get benefit from each other.

Benefits of Social Organization

Benefits of Social Organization

2. Sociality offers **several benefits to animals reproduction**. It facilitates encounters between males and females, which, for solitary animals, may consume much time and energy. Sociality also helps synchronize reproductive behavior through the mutual stimulation that individuals have on one another.

Examples:

- Large colonies of gulls produce more young per nest than do small colonies. Furthermore, parental care that social animals provide their offspring increases survival of the brood.

Benefits of Social Organization

Benefits of Social Organization

3. There is **improved protection of limited recourses** against other groups of the same species and related species. Members help each other in protecting territory, food material and mates against other groups. Social living provides opportunities for individuals to give aid and to share food with young other than their own. Such interactions within a social network have produced some intricate cooperative behavior among parents, their young, and their kin.

Benefits of Social Organization

Benefits of Social Organization

4. There are increased chance of foraging success. In other words more individuals finding food is better than one.

Examples:

All the lions in the pride cooperate with each other in foraging and protection, though there is still serious competition among the males. Male lions work together to defend the other lions in the pride. Female lions work together to hunt.

Benefits of Social Organization

Benefits of Social Organization

3. There is reduction in predation pressure by improved detection and repulsion of enemies.
4. There is more care of offspring through communal feeding and protection.

Examples:

An infant yellow baboon *(Papio cyanocephalus)* “jockey rides” its mother. Later, as the infant is weaned, the mother-infant bond weakens and the infant will be refused rides.

Benefits of Social Organization

Benefits of Social Organization

5. Of the many other advantages of social organization, cooperation in hunting for food; huddling for mutual protection from severe weather; opportunities for division of labor, which is especially well developed in the social insects with their caste systems; and the potential for learning and transmitting useful information through the society.

End

Lecture # 56

56. Disadvantages of Social Organization

Disadvantages of Social Organization

Disadvantages of Social Organization

There are some disadvantages of living in social organization.

Few of them are discussed here:

1. There is increased competition within the group for food, mate, nest sites, nest material and other limited recourses.
2. Increased risk of infection due to outbreak of disease or epidemic.
3. Increased risk of exploitation of parents.
4. Increased inbreeding will weaken the progeny.

Benefits of Social Organization

- **Increased Sickness and Disease**

Animals that live in close proximity to one another face higher risks of infection than do individual animals.

Diseases and parasites can be passed between animals living too closely together, and this can dramatically decrease a group's numbers.

Some animals might die as a direct result of the sickness itself, but latent effects of infection can come in several forms, such as decreased mobility, making it difficult to escape from predators, as well as decreased vision and sense of smell, making it harder to find food.

Benefits of Social Organization

- **Increased Competition for Food**

The bigger the animal group, the more competitive the hunt for food becomes.

Predators who hunt in large packs or prides tend to expend more energy than needed in capturing prey, as the element of surprise is not so easy to come by.

Predators often scavenge as a result, and many go hungry. Sizable assemblies of animals put pressure on food resources, and unequal distribution of food among animals can lead to increased aggression, starvation, and death.

Benefits of Social Organization

- **Increased Competition for Mates**

In group, where there are more males than females; direct, aggressive competition to find a mate can become the norm.

Ritualized fighting and displays of violence and aggression can ensue before and even after finding a mate. The ultimate battle here is not necessarily for survival in the strictest sense but in the passing on of genes and reproductive success.

End

Lecture # 57

57. Social Organization in Insects

Social Organization in Insects

Social Organization

- All species show some degree of social organization; many species nest in a system of tunnels, in the soil, under a dome, hill or excavated earth, sand, or debris.
- Colonies range in size from a few dozen to half a million or more individuals. Typically they include three castes: winged, fertile females, or queens; wingless, infertile females, or workers; and winged males.

Social Organization in Insects

Social Organization

- The familiar example are of social social organization in insects are:
  - ants, bees, and wasps(order hymenoptera),
  - termites (order isoptera),
  - Silkworms

Many of these species live in colonies with thousands or even millions of individuals.

Social Organization in Insects

Ants

- The most highly developed social insects.
- Each colony contains at least one queen. Ants defend their nest by biting and stinging.
- Ants secrete pheromones detected by other ants through antennae. This process called **chemoreception;** the primary communication vehicle that facilitates mate attraction, kin and non-kin recognition.
- Ants send tactile signals by touching and stroking each other's bodies with their antennae and forelegs.

Social Organization in Insects

Ants

- Majority of ants form underground nests called **formic aria.** It consists of numerous chambers connected by galleries. The Chambers are used as granaries for storing seeds for rearing progeny.
- Ants fall into four main castes:
  - **QUEEN:** Fertile females with reproductive system. They are larger than all other castes. Two pairs of membranous wings and mandibles are present. Their sole duty is to lay eggs. They may have life span of 15 or even 30 years. The ant colony has several queens.

Social Organization in Insects

Ants

- **KINGS:** Fertile males with well reproductive system, smaller than the queens and soldiers but larger than workers. They have two pairs of wings and mandibles are reduced. They serve to fertilize the queens.
- **WORKERS:** These are sterile females with reduced, non functional reproductive system. They are smaller than all other castes. Their mandibles are well developed but eyes are small. They repair nest ,collect and store food.

Social Organization in Insects

Ants

- **SOLDIERS:** They are modified workers. They are smaller than queen in size. They are wingless with large heads and powerful mandibles. They defend the colony from the enemies. They also crush seeds and fruits.

Social Organization in Insects

Termites

- They are social insects but less advanced that ants colonies.
- The termite colony has three class of individuals and each class includes both sexes.
- Termites digest wood and paper. Some species of termites build nests up to 6 meters in height.

Social Organization in Insects

Termites

- Termites use pheromones for communication.
- They use pheromones for trail, alarm, aggregation and mating.
- Trail pheromones is produced by termites when they are trying to get other member of the colony to follow then to a particular area perhaps a new food source for the colony. Termites dislike daylight. Some species lives in nests called **termitaria.**

Social Organization in Insects

Termites

Termites are categorized as:

- Fertile castes
- Sterile castes

Social Organization in Insects

Termites

- **Fertile castes** also known as sexual castes. It is of 3 types:
  - **LARGE WINGED FORMS:** These are king and queen. The sole duty of queen is to lay eggs.
  - **SHORT WINGED FORM:** These are supplementary kings and queens. The wings are short, pad like and vestigial.
  - **WINGLESS FORM:** Wings are absent. The ocelli are absent.

Social Organization in Insects

Termites

- **Sterile Castes:** These are wingless, eyeless and colorless. The sex organs are rudimentary. It is of 3 types:
  - **WORKERS:** They are dimorphic. They are trimorphic in some cases. They construct, repair the nest collect and distribute food.
  - **SOLDIERS:** They are larger than the workers. The soldiers defend the colony by killing the intruders with their mandibles.
  - **NASUTES:** The nasutes also called the proboscideans. They possess large head prolonged into rostrum.

Social Organization in Insects

Honey bee

- They lives in hives which is often constructed in a hollow tree. The individuals forming a colony are of 3 castes:
  - **QUEEN:** The queen is fertile female. She is unable to form hive. Her duty is to lay eggs.
  - **DRONES:** The drones are males. They do not work in hives.
  - **WORKERS:** They are sterile females. Young workers stay in the hives and called the house bees.

Scout bees fly from the colony in search of pollen and nectar. It successful in finding good supplies of food “dances” on the honeycomb.

Social Organization in Insects

Yellow wasp

- Also known as **Polistes**. The wasp has 3 castes: **queens, drones and workers**.
  - Single **queen** prepares a nest in spring. The queen lay eggs and the egg hatch into apodous larvae .
  - The **workers** have a powerful sting and are a source of great annoyance to men.
  - Drones are produced from unfertilized eggs. They fertilize the young queens.

In winters the young queens hibernate and the nest perishes .The queens start new colonies next spring. The adult wasp feed on insects ,fruit juices and nectar.

End

Lecture # 58

58. Social Organization in Birds

Social Organization in Birds

Social Organization

The term social organization includes:

- **Physical structure:** the size of the group and its composition in respect to age, sex and degrees of relatedness of group members.
- **Social structure:** all of the relationships among individuals in the group and their consequences for spatial distribution and behavioural interactions.
- **Group cohesion:** the duration of association of the members of the group and the frequency of fission in which one or more members leave the group.

Social Organization in Birds

Social Organization in Chicken

- There is also a social organization in chicken called the **pecking order**. It is a system by which birds arrange their social standing in the flock.
- The higher ranked birds will get the best food, water and roosts while the lower placed birds will get the leftovers.
- This method of organization places each member of the flock on a **'hierarchy ladder'.** At the top of the ladder will be the **head rooster** (or hen if no rooster is present).

Social Organization in Birds

- It is a flexible structure and within the flock there are usually three different types of social order going on:
  - Rooster to Rooster
  - Hens to Hens
  - Roosters to Hens
- A **rooster** may go up the ladder if he mounts a successful campaign against the leader. He becomes the new chicken-in-charge! And the defeated roosters go down the ladder, as do weak or sickly birds.

Social Organization in Birds

- Hens have their own order. The matriarchs of the flock will be up to the top of the ladder, with less dominant birds at the bottom.
- In this system the older, stronger and savier hens will be at the top.
- **Young pullets** just coming to point of lay, will try to ‘move up’ the social ladder quickly.
- If a hen tries to challenge; she will earn glares, pecks and feather pulling from the higher ‘ranked’ hens.

Social Organization in Birds

- The serious games of the pecking order start when chicks are around six weeks of age.
- Chicks will start rushing at each other, bumping chests and flaring feathers. These are all methods used to intimidate flock mates at any stage of life. By the time they leave the brooder, they will have their own pecking order sorted out.

Social Organization in Birds

Social Organization in Starling

- Starlings normally fly in loose flocks, but when under attack by a peregrine falcon, they fly as close together as possible.

Social Organization in Birds

Social Organization in Pied Flycatcher

- A male pied flycatcher defends a territory against other males. The female chooses a male on the basis of the quality of his territory and the suitability of the nest site it contains. Males that had a low density of birch trees in their territories and nest sites high up in the trees with thick trunks giving safety to the nest were the ones that obtained mates first.

Social Organization in Birds

Nesting and Parental Care in Birds

- 95% of bird species are socially **monogamous** i.e., a male and female bird will pair for at least the length of the breeding season or, in some cases, for several years or until the death of one mate.
- Monogamy allows for care of the fledglings by both parents.

Social Organization in Birds

Nesting and Parental Care

- Some birds are polygamous; they have many mates in a breeding season.
- Birds eggs are usually incubated by the parents. Most birds have an extended period of parental care after hatching. The length and nature of parental care varies widely amongst different orders and species.

End

Lecture # 59

59. Social Organization in Mammals

Social Organization in Mammals

Social Organization in Mammals

Social organization can also be observed in variety of mammalian species.

Examples:

Black Tailed Prairie dog

1. Highly social, social units are called **Coteris**.
2. Coterie is composed of one adult male, several females and young ones.
3. Without dominant hierarchy.
4. Everything inside the premises of coterie are shared by all members.

Social Organization in Mammals

5. Hostility with neighbors always exist.
6. Two syllable call used to claim the territorial ownership.
7. They become familiar to each other and start recognizing a member of same territory.
8. In a coterie, dog watch out for danger, produce alarm calls.

Social Organization in Mammals

Examples:

Meerkats

1. Active and social animals
2. Live in groups that include 2-30 individuals, although the average pack size is around 10-15 individuals. Groups are called **mobs**.
3. Each mob may consist of up to three families living together. Each family group consists of a breeding pair and their offspring.

Social Organization in Mammals

4. Each mob comprise on nearly equal numbers of either sex.
5. Members take turns at jobs such as looking after pups and keeping a lookout for predators.
6. Meerkats are a cooperatively breeding species — typically the dominant **'breeders'** in a mob produce offspring, and the nonbreeding, subordinate **'helpers'** provide altruistic care for the pups.
7. Meerkats have a clear dominance hierarchy with older individuals having a higher social status.

Social Organization in Mammals

Examples:

Elephants

1. Several levels of social system can be observed in elephants: **Family, kin and Bachelor's herd.**
2. Mostly 10-20 members are there.
3. Family consists of grandmothers, daughters, sons, grandsons and grand daughters.

Social Organization in Mammals

4. Females bond socially for nearly 50 years.
5. Kinship involves several family groups living in same vicinity, occasion and mingle peaceably. They are not together except during the migration times.
6. At their maturity, bulls leave their family and join Bachelor herds, all male herd. It is dominated by ritual dominance , and dominant male gets a female in estrus.

Social Organization in Mammals

Examples:

Primate societies

Variety of social organization exists in primates. The group could comprise of one male and a group of females called **harem**, a multi male group or a large **troupe.**

1. **Solitary ones** are the **lemurs, loris and tarsiers**.

   They are arboreal, nocturnal and feeding on insects. They make contacts during the breeding season.

Social Organization in Mammals

2. **White handed gibbons** are monogamous species. A pair lives with their offspring. The offspring move away after they become sexually mature. They are territorial.
3. The **gelada baboons** show variation in social organization depending on the availability of food. There are single male herds. There are herds with “single” male and a “harem” of females.

Social Organization in Mammals

4. The **mountain gorilla** lives in small groups, headed by a single male and includes other males as well. They are not territorial.
5. Some of the smallest primate societies are seen in the gibbon and orangutan. **Orangutan** is an exceptionally solitary animal while the **gibbons** live as small family units, the father, mother and their offspring. Some of the largest groups are to be found in the terrestrial primates-baboon troops range up to 90.

End

Lecture # 60

60. Territoriality and Defense

Territoriality and Defense

Territoriality

- A **territory** is an area defended against other members of the same species. It provides food, water supplies, nesting areas and refuges from danger.
- Territoriality uses a great deal of an individual's energy. In addition, an individual might die defending a territory, thus miss a reproductive opportunity.

Territoriality and Defense

Examples of Territorial Behaviour

- Spraying behavior is a way for an individual to mark its territory.
- The red breast of the robin (visual)
- Bird songs or the calls of gibbons (auditory)
- Through the deposit of scent marks (olfactory)
- Bears, dogs, cats, lemurs, etc. use scent-marking to signal the boundaries of their territories.

Territoriality and Defense

Territories are usually used for:

1. Feeding
2. Mating
3. Rearing of young
4. Combination of all activities

Territoriality and Defense

Types of Territories

- **Mating Territory:** places of mating for the partner.
- **Breeding Territory:** The breeding territory is relatively small. It usually contains only a nesting or mating site.
- **Feeding Territory:** It tends to be larger than the breeding territory because it must contain sufficient food to support the owner of the territory and any mate or offspring that may also be residing there.
- **Multi- purpose Territory:** This type of territory is generally the largest as it includes aspects of both breeding and feeding territories.

Territoriality and Defense

Salient Features of a Territory

1. Definite home and home range

Territorial behaviour varies widely. Most animals have a **definite home.** The area the animal covers regularly in search of food and mates is the **home range**. This area is not defended. The part of the home range defended against others of the same species is the territory. Aggressive behaviour is used to hold on territories.

Territoriality and Defense

2. Adaptive Features of Territoriality

Ensures enough space for each animal – if in short supply and needed for breeding, keeps population down.

By spreading out reduces the spread of disease and parasites. Also harder for predators to find them.

Most successful males hold best territories and so ensure best genes are passed on to offspring. Once territories are established the resources have been divided. The losers will spread out and look for food elsewhere rather than go on fighting.

Territoriality and Defense

3. Food and security

Territories ensure enough food for the animals and their families. Territories ensure a safe, protected nest or home for the young or at least a place to breed in the case of communal breeding grounds. Animal now has an area with which it can become familiar, can learn where food, water and protection from predators is located. Territorial behaviour is set. Defenders and intruders know their roles.

Territoriality and Defense

4. Marking and Defending Territories

- **Vocalizations** – e.g. birds singing on boundaries of their areas at dawn and dusk
- **Scent** – e.g. marking with urine (dogs and cats) or faeces
- **Scent glands** – special glands produce chemo markers. e.g. on rump, between horns (deer), wrists (lemur), behind ears (cats)
- **Physical gesturing** – crabs wave claws at edge of territory

Territoriality and Defense

Advantages of territory

Territorial behavior forms an important part of social behavior.

- ✓ Territory provides a base which the male can advertise himself to female, and attracted the female.
- ✓ Provides a protected place for courtship, mating and parental care to young.
- ✓ Provided safety where young can live freely and without any interference.

Territoriality and Defense

Defensive Behaviour

- Behavior showed by animals trying to avoid being harmed by another.
- This might involve:
  - ✓ **defensive aggression:** against a potential predator or against a member of its own species, from its own group or an outside group
  - ✓ **animal escape behavior** to avoid the source of harm
  - ✓ **Camouflage** or inactivity
  - ✓ highly conspicuous, as in the noise that a rattlesnake makes to indicate potential danger if provoked or attacked

Territoriality and Defense

Types of Defensive Behaviour

1. Chemical defense

Examples

- **Poison dart frog** has poison glands secreted all over its body.
- **Fire salamander** makes a nerve poison, which it is secreted by the glands on its back.
- **Bombardier beetle** spray boiling hot and chemically toxic bodily fluids towards predator.
- **Horned lizard**, when attacked; it pressures its own sinus cavities until the blood vessels in its eyes burst, then sprays its attacker with blood from its eyes.

Territoriality and Defense

Types of Defensive Behaviour

2. Camouflage

Examples

- Stick insect pretends to be a twig, in an attempt to avoid being seen by a bird or other predator.
- Polar bear blend into the white snow of the Arctic to avoid danger.
- Frogfish resembles a sponge and prey on the small fishes swimming nearby.
- Chameleon only change color when in imminent danger.

Territoriality and Defense

Types of Defensive Behaviour

3. Actual Weapons

Examples

- The simple defense system of turtles is its shell. Predators cannot penetrate it.
- Hedgehogs have spikes that are enough to deter predators.
- Pufferfishes have the ability to inflate themselves into a balloon shape by swallowing water. Their bodies are also covered with thorny spikes, which stand out on inflation, providing a formidable defense against attack.

Territoriality and Defense

Types of Defensive Behaviour

4. Mimicry

Examples

- The simple defense system of turtles is its shell. Predators cannot penetrate it.
- Scarlet king snake is non-poisonous and mimic of the extremely venomous coral snake.
- Locust borer not only looks like a bee or wasp, it sounds like one, too.

End

Animal Behavior

Lecture # 61

Social Behaviors

Social Behaviors

Social Behaviors in Animals

- A set of interactions among individuals of the same species
- A wide range of sociality occurs among animals

Social Behaviors

Asocial Animals

- Some animals rarely interact with one another e.g.
  - Mosquitoes
  - Polar bears

Social Behaviors

Social Animals

- Social animals live together in large groups or form tightly knit colonies, e.g.
  - Ants and termites
  - Bees and wasps
  - Many birds
  - Wildebeests
  - Wolves and Lions
  - Monkeys and gorillas
  - Humans

Social Behaviors

Characters of Social Behaviors

- Most social behaviors are based on
  - Cooperation
  - Competition
  - Conflict
  - Exchange
  - Coercion
  - Sacrifice
- Most social animals cooperate to accomplish some tasks

Social Behaviors

Characters of Social Behaviors

- Many social behaviors are adaptive
- They increase animal's fitness and reproductive success
- Animals in groups show joint aggregation against predators

Social Behaviors

Characters of Social Behaviors

- Many social behaviors are agonistic and based on competition
- Such behaviors are selfish and individuals benefit at the expense of others

Social Behaviors

END

Characters of Social Behaviors

- Most social adaptations benefitting one individual may indirectly harm others, e.g.
- Superior foraging ability of one individual may leave less food for others
- Some behaviors are altruistic or unselfish i.e. an animal favors other sacrificing its own benefit

Animal Behavior

Lecture # 62

Social Behavior:
Living in Groups

Social Behavior: Living in Groups

Animal Groups

- Animal populations are often organized into groups
- Groups may be:
  - Simple aggregation of individuals
  - Complex animal societies

Social Behavior: Living in Groups

Aggregation

- An aggregation is any form of gathering of animals or the process of coming together
- There is minimal interaction between members of an aggregation
- For example, several *Drosophila* flies living on a piece of rotting fruit

Social Behavior: Living in Groups

Animal Societies

- A stable group of individuals of a species
- It maintains a close-knit cooperative social relationship b/w individuals
- It has a complex social organization
- Its sphere extends beyond the level of mating and taking care of young

Social Behavior: Living in Groups

Advantages of Social Grouping

- Protection from predators
- Warning signals spread rapidly
- Increased feeding efficiency by cooperative hunting and food search
- Protection from harsh cold environment by social huddling
- Helpful in finding mates and rearing of young

Social Behavior: Living in Groups

Disadvantages of Social Groupings

- Competition for resources (food, mates) develops b/w members of a social group
- Diseases and parasites spread more rapidly
- Interference in reproductive behaviors and rearing of young
- Increased susceptibility to predation for large congregations of animals

END

Animal Behavior

Lecture # 63

Altruism

Altruism

Altruism

- It is a type of social behaviour
- A selfless behavior in which an individual endangers his own survival or sacrifices some of its own reproductive potential to benefit another individual.
- It is an innate trait with genetic basis
- Developed through the process of evolution in many animal species

Altruism

Altruism

- Particularly prevalent in species with complex social structures e.g. primates, some other mammals and insects
- There are wide individual variations in tendencies toward altruism

Altruism

Altruism

- Altruism is a different kind of social behavior
- It has been hard to explain within the framework of Darwinism and natural selection which emphasize on fitness
- It reduces the fitness of the altruist individual

Altruism

Altruism

- Two basic concepts comes under Altruism:
  - Inclusive fitness
  - Kin selection

Altruism

Altruism

- **Inclusive fitness** is defined as the affect an individual has on proliferating its own genes by reproducing and helping relatives raise offspring.
- **Kin selection** is the mechanism of inclusive fitness, where individuals help relatives raise young.

Altruism

Significance of Altruism

- In ethology, costs and benefits of a behavior are measured in terms of reproductive fitness, or expected number of offspring
- By behaving altruistically, an organism reduces the number of offspring it is likely to produce itself

Altruism

Significance of Altruism

- By sacrificing its own favors to save others, an altruist indirectly boosts the would-be number of offspring of species—that other members are likely to produce.

End

Animal Behavior

Lecture # 64

64. Examples of Altruism

Examples of Altruism

Altruism in squirrel

- A squirrel on seeing a predator gives alarm calls to alert unaware individuals to hide safely
- Squirrel that warns becomes conspicuous increasing risk of its own predation
- Similar altruist alarming calls are observed in crows and monkeys

Examples of Altruism

Altruism in social insects

- Social insects such as ants, wasps, bees and termites also shows altruism.
- In social insect colonies, sterile workers devote whole lives to:
  - Care for queen
  - Construct and protect the hive
  - Foraging for food
  - Tending the larvae
- This is maximally altruistic behavior.

Examples of Altruism

Altruism in stinging bees

- Worker bee stings intruders to help defend the hive—but result is death of stinging bee

Examples of Altruism

Altruism in Vampire bats

- Vampire bats regurgitate blood and donate it to other members of their group who have failed to feed that night, ensuring they do not starve

Examples of Altruism

Altruism in Emperor penguins

- Emperor penguins migrate miles in harsh conditions to bring back food for their young.

Examples of Altruism

Altruism in Vervet monkeys

- Vervet monkeys give alarm calls when predator approaches. Alarm calls attract attention of predator and others monkeys have more time to escape.

End

Lecture # 65

Kin Selection and Inclusive Fitness

Kin Selection and Inclusive Fitness

Kin Selection

- An altruistic evolutionary strategy favoring reproductive success of an animal's close relatives—even at a cost of animal's own survival and reproduction

Kin Selection and Inclusive Fitness

Kin Selection

- Kin selection is an instance of inclusive fitness
- It combines the number of offspring produced by an individual itself and the number of offspring it can ensure by supporting other close relatives (siblings)

Kin Selection and Inclusive Fitness

Kin Selection

- Fitness of an individual is based on the genes it passes on as well as on those common genes that its relatives pass on
- Through kin selection, a gene carried by a particular individual passes on to the next generation through a related individual

Kin Selection and Inclusive Fitness

Kin Selection

- Kin selection explains how natural selection acting on relatives of an animal can affect the fitness of an individual
- For kin selection to be effective, individuals of a group must be able to identify their relatives
- So, it is found in animals with advanced social structures i.e. primates and social insects

Kin Selection and Inclusive Fitness

Kin Selection: Example

- Altruistic kin selection behavior is most evident in sacrifice of parents for their offspring
- Parents' sacrifice of their own reproductive potential actually increases their fitness because reproduction by their kins maximizes spread of their genes in population

Kin Selection and Inclusive Fitness

Inclusive Fitness

- Inclusive fitness is the measure of fitness of a group of members of a species—which may not be close relatives

Kin Selection and Inclusive Fitness

Inclusive Fitness

- An organism's inclusive fitness is determined by its personal fitness, plus its weighted effects on the fitness of every other individual in population
- Weights are determined by relationship coefficient (related genes)

Kin Selection and Inclusive Fitness

Inclusive Fitness

- It is favored by natural selection
- N.S. acts to maximize genetic and reproductive fitness of group (population)
- It does not think in terms of maximizing an individual's genes in the gene-pool

Kin Selection and Inclusive Fitness

Inclusive Fitness Behavior

- Individuals of a species helping others who are not their close relatives

END

Lecture # 66

Reciprocal Altruism

Reciprocal Altruism

Reciprocal Altruism

- Animals occasionally behave altruistically toward unrelated individuals
- This is in expectation of return of favor in future—i.e. reciprocity
- A baboon may help an unrelated companion in a fight
- A wolf may offer food to another wolf—even they share no kinship

Reciprocal Altruism

Reciprocal Altruism

- Reciprocal altruism is limited to species with stable social groups (e.g. chimpanzees, humans)
- In such groups, individuals are likely to meet again and get chance to exchange aid

Reciprocal Altruism

Cheating

- Reciprocal altruism is also associated with negative consequences
- If favors are not returned to individual who had been helpful in past
- Such unreciprocated behavior is referred to as “cheating” by behavioral ecologists
- It occurs as cheating may benefit the cheater substantially

Reciprocal Altruism

Game Theory and Tit for Tat Strategy

- Game theory explains how reciprocal altruism dominates over cheating behavior
- It involves a behavioral strategy: tit for tat
- In this strategy, an individual treats other in the same way it was treated in their last encounter

Reciprocal Altruism

Game Theory and Tit for Tat Strategy

- Individuals adopting tit for tat behavior are always altruistic, or cooperative, on the first encounter with another
- They remain so as long as their altruism is reciprocated

Reciprocal Altruism

Game Theory and Tit for Tat Strategy

- When their cooperation is not reciprocated, individuals employ tit for tat—retaliate immediately
- Cooperative behavior returns as soon as other individual becomes cooperative

END

Animal Behavior

Lecture # 67

Examples of Reciprocal Altruism

Examples of Reciprocal Altruism

Reciprocal Altruism in Cleaner fish

- Cleaner fish and their clients include other large fishes.
- The host fish allows the cleaner fish free entrance and exit and does not eat the cleaner, even after the cleaning is done.

Examples of Reciprocal Altruism

- The host signals the cleaner it is about to depart the cleaner's locality, even when the cleaner is not in its body. The host sometimes chases off possible dangers to the cleaner.

Examples of Reciprocal Altruism

Reciprocal Altruism by Red-winged blackbird

- Red-winged blackbird males help defend neighbor's nests.
- There are many theories as to why males behave this way.
    1. Males only defend other nests which contain their extra-pair offspring.

Examples of Reciprocal Altruism

2. Another is the tit-for-tat strategy of reciprocal altruism.
3. A third theory is, males help only other closely related males.

The tit-for-tat strategy of reciprocal altruism is highly accepted by researchers.

Examples of Reciprocal Altruism

Reciprocal Altruism in Primates

- Grooming and coalitions in female vervet monkeys is an example.
- Human social behavior also shows the reciprocal altruism; explains human emotions and ethics etc.

Examples of Reciprocal Altruism

Reciprocal Altruism in Wolves

- Wolf also shows reciprocal altruism by helping out other wolves that are not even the members of their pack.

Examples of Reciprocal Altruism

Reciprocal Altruism in dolphins

- There are cases reported about the reciprocal altruism by dolphins.
  - The surfer, Todd Endris was surfing at Marina State Park off Monterey, California, when he was suddenly attacked by a great white shark.
  - The shark peeled the skin off his back and was biting on his right leg.
  - A pod of bottlenose dolphins swam over and formed a protective ring around Endris, providing him a chance to escape from the shark and to swim back to shore where help was available.

Examples of Reciprocal Altruism

Reciprocal Altruism in dolphins

- Moko, a bottlenose dolphin in New Zealand, helped two whales who were stranded on the beach by directing them to open sea.

End

Lecture # 68

Parental Care

Parental Care

Parental Care

- Catering the needs of newborn and young offspring by parents
- More evolved in animals whose newborns and young cannot care for their selves
- This behavioral need has been an important factor in the evolution of mating systems

Parental Care

Parental Care in Birds

- Birds show maximum parental care
- Most newly hatched birds cannot care for themselves
- They require a large, continuous food supply

Parental Care

Parental Care in Birds

- Catering the needs of young is difficult for a single parent
- To enhance chances of viability of offspring, most birds form monogamous pairs
- Male stays with its mate and helps her till offspring are able to fulfill their own needs

Parental Care

Parental Care in Mammals

- Lactating female nourishes the young
- She provides most of the required care
- Males usually play no role in raising the young

Parental Care

Parental Care in Mammals

- In some species (e.g. lions) males provide protection to females and young
- In such species, a male or small group of males takes care of many females at once in a harem

Parental Care

Polygyny Prevails If No Parental Care Required

- Males in mammals and birds do not stay with their partner if young ones can feed and care for themselves after hatching
- Males of these species maximize their reproductive success by seeking other mates
- Polygyny is more common in such animals

Parental Care

Male Parental Care: Certainty of Paternity

- Mating behavior and parental care of males is associated with certainty of paternity
- Certainty of paternity is relatively low in most species (even within a monogamous relation)
- So, male parental care is found in very few species of birds and mammals

Parental Care

Male Parental Care: Certainty of Paternity

- Males engage in behaviors that increase their certainty of paternity—including:
- Guarding females
- Removing any sperm from female reproductive tract before copulation
- Introducing large quantities of sperm to displace sperms of other males

In such species

END

Animal Behavior

Lecture # 69

Examples of Parental Care

Examples of Parental Care





Salmon and trout

Parental Care in fishes

- The simplest form of parental care is guarding or protection of eggs in egg-laying, or oviparous, species.
- For example female salmon and trout will excavate nests by digging simple depressions with their tails. The eggs that are laid in the nests are fertilized and then buried by the female.
- Brood fishes carry their eggs in their mouth to keep them safe from predators.

Examples of Parental Care



Green frog



Blacksmith frog

Parental Care in Amphibians

- Paternal care occurs in a number of species of anuran amphibians.
- Males of green frog defend their eggs by not allowing small sized intruders in their territories.
- Blacksmith frog digs small holes in the mud for deposition and development of the eggs.

Examples of Parental Care



Aquatic salamanders



Japanese giant salamander

Parental Care in amphibians

- Aquatic salamanders, females after laying eggs guard them by coiling body till the eggs hatch.
- In Japanese giant salamander, the males perform the same function.

Examples of Parental Care

Parental Care in reptiles

- Reptiles provide less parental care than other tetrapods. When it does occur, it is usually female-only or bi-parental care.
- A few lizards and snakes guard their eggs and pythons incubate their eggs for a while.
- The eggs of some female snakes remain inside the body until they hatch but also receive no parental care.

Examples of Parental Care

Parental Care in birds

- 96% of birds show parental care to their offspring.
- Bi-parental care is the most common form in birds, especially in **passerines**. The hatchling benefits from the mutual care at the cost of the parents' future reproductive success.
- In bi-parental care, the male provides food and the female is a caretaker. Examples include acorn woodpeckers and red-eyed vireos

Examples of Parental Care



Snail kite

Parental Care in birds

- **Snail kites** are example of mono- parental care in birds. The both members compete on who will desert the nest.
- In some species females never incubate offspring alone unless the male has been killed for example spotted sandpipers and red-necked phalaropes.
- In temminck's stint, little stint, mountain plover, and sanderling the females lays a clutch of eggs and the male incubates them.

Examples of Parental Care

Parental Care in Mammals

- There is maternal care in all species of mammals. 95% species exhibit female-only care. In the remaining 5% of species, there is bi-parental care.
- In **rodents**, parental care involve behaviors as warming, feeding, protecting, retrieving, and grooming young depending on the species.
- In domestic **dogs**, maternal behaviour shown by the dam mainly consists of contact, nursing, grooming/licking, play, punishment, thermoregulation, and motion.

Examples of Parental Care

Parental Care in Mammals

- Among **primate** species, male also involve in parental care. This care is often in the form of grooming, carrying, support in agonistic interactions, and protection against infanticide.
- Owl monkeys (Aotus azarai) and Titi monkeys (Callicebus caligatus) exhibit bi-parental care for their offspring.

Examples of Parental Care

Parental Care in Mammals

- Mother **bats** hold the pups in their tail pouches. Pups are only an inch long and suck milk from their mothers.
- In **cetaceans**, maternal care is long-lasting. Calves and juveniles stay close to their mothers‘ sides for a long time, even if they are able to feed themselves.
- **Belugas** remain near their mothers for three years, sometimes even longer.
- The rearing and socialization of **bottlenose dolphins** calves depend on the female, whether via direct maternal care or allomaternal parenting.

End

Lecture # 70

Social Learning

Social Learning

Social Learning

- “Type of learning that involves observing and copying others behavior”
- Example: young chimpanzee learns to crack palm nuts by copying the behavior of elders

Social Learning

Social Learning and Culture

- Social learning forms the roots of culture
- Culture: “a system of information transfer through social learning or teaching that alters behavioral phenotypes of individuals in a population”

Social Learning

Social Learning and Culture

- Culture enhances the fitness of individuals much rapidly
- Culture based changes in phenotype occur on a much shorter time scale than changes resulting from natural selection

Social Learning

Social Learning Behaviors

- Social learning behaviors are prevalent in animal kingdom, specially vertebrates
- Types:
  - Copying behaviors
  - Fine-tuned social learning

Social Learning

Mate-Choice Copying Behavior

- A form of social learning behavior demonstrated in several fish, bird and mammal species
- Individuals in a population copy the mate choice of others
- In copying mate choice, an individual is attracted to a mate that is also attractive to most other individuals

Social Learning

Fine-Tuned Social Learning

- Fine-tuned learning with experience in humans is well known
- A human baby starts to learn expressive words and accents that modify and improve with age
- Studies on vervet monkeys have demonstrated fine-tuned learning in animals

Social Learning

Fine-Tuned Learning of Alarm Calls

- Vervet monkeys produce a complex set of alarm calls
- They give distinct alarm calls for predators like leopards, eagles or snakes
- Alarm call for leopard is a loud barking sound
- For an eagle a short double-syllabled cough
- For a snake, a “chutter”

Social Learning

Fine-Tuned Learning of Alarm Calls

- Upon hearing an alarm call, other vervets behave in an appropriate way:
- Run up a tree on hearing alarm call for leopard
- Look up on hearing alarm for an eagle
- Look down on hearing alarm for a snake

Social Learning

Fine-Tuned Learning of Alarm Calls

- Infant vervets give and understand alarm calls, but in a relatively undiscriminating way
- They give “eagle alarm” on seeing any bird
- With age, they fine-tune their calls by learning from other members of the group

END

Animal Behavior

Lecture # 71

Examples of Social Learning

Examples of Social Learning

Social Learning in birds

- A bird is able to learn by **observation and imitation** of others.
- Bird song is a good example of social learning. Bird song is affected by biological factors such as hormones and maturation, birds in different regions have different "dialects".

Examples of Social Learning

Social Learning in birds

- Birds learn **to solve complex problems** more readily than many mammals in laboratory experiments.
- Birds learn how to capture food.
- Blue Jays can learn to tell edible from inedible caterpillars by watching the reaction of other Blue Jays when they attempt to eat a caterpillar.

Examples of Social Learning

Social Learning in birds

- A number of birds are able **mimics**. For example a **parrot** can be taught to say different phrases of human language.
- A **Northern Mockingbird** can mimic as many as 30 other species of coexisting birds.
- The **Marsh Warbler** in mimics 99 European species and 133 African species that it coexists with on its wintering grounds.

Examples of Social Learning

Social Learning in birds

- A **Lyrebird** learned to imitate barn yard sounds including pig squeals, a chain rattling, a cross-cut saw, and a howling dog, along with most calls and songs of the local birds.

- A **Crested Lark** learned the whistled commands given to sheepdogs (“Run ahead”, “Fast”, “Halt” and “Come here”). When taped and played back to sheep dogs, these calls were obeyed by the dogs.

Examples of Social Learning

Social Learning in birds

- Some birds have devised ways to **use tools to feed**.
- The **Woodpecker Finch** of the Galapagos Islands uses a stick or cactus spine held in its bill to pry insects from crevices.
- **Egyptian Vultures** have learned to crack ostrich eggs using stones.

Examples of Social Learning

Social Learning in birds

- **Green Herons** are clever fishers, they use pieces of bread as fishing bait. A Green Heron will drop bread into still water and wait for fish to gather which it then captures by spearing with its bill.

Examples of Social Learning

Social Learning in Fishes

- Fish use social information in a variety of contexts, including: antipredator behavior, migration, foraging and mate choice.

- Fishes shows fright responses. When one member of a group is startled, the rest of the shoal respond to it. This reaction can be driven both by visual information and chemical alarm substance in fishes of the superorder Ostariophysi.

Examples of Social Learning

Social Learning in Fishes

- Naive fish learn to associate a new predator with the visual fright response of more experienced fish.

Examples of Social Learning

Social Learning in other animals

- Young chimpanzees learn to use tools to open nuts by observing the adults.
- Chimpanzees learn to solve stick and tunnel problems, ant species selection to eat and also able to learn sign language in experimental setups.

Examples of Social Learning

Social Learning in other animals

- Young rhesus monkey (*Macaca mulatta*), learns to fear snakes by observing their parents being afraid of them.
- Orangutan learns to dig with spade, lift lid of sewage tank, building nests, feeding habits, flaking stone tools and tool use during experimental studies.

Examples of Social Learning

Social Learning in other animals

- Adult **meerkats** (*Suricata suricatta*) teach its pups the essential prey-handling skills.
- Meerkats feed on dangerous prey (such as venomous scorpions) that inexperienced pups appear unable to safely catch and eat.

Examples of Social Learning

Social Learning in other animals

- Adults display teaching behavior to pup and modify their specific teaching behaviors based upon the age of the pup.

- Teaching behaviour is also seen in other species. Cheetah catches live prey animals and transport them back to cubs, allowing the cubs to learn and practice hunting skills.

End

Lecture # 72

Evolution and Human Culture

Evolution and Human Culture

Genetic Basis of Human Culture

- Human culture and certain social behaviors are due to expression of genes
- Genes perpetuate by natural selection
- So, human culture is related to Darwinian evolutionary theory

Evolution and Human Culture

Genetic Basis of Human Culture

- An irrational interpretation of genetic and evolutionary basis of human behavior may be misleading
- It will justify the status quo in human society, rationalizing social injustices and animal behaviors of human

Evolution and Human Culture

Plasticity of Human Behavior

- Human behavior is not a rigid genetic model
- Just as anatomical features vary extensively, so do inherent behaviors
- Just like environment intervenes in phenotypic expression of physical traits—it affects the expression of behavioral traits

Evolution and Human Culture

Plasticity of Human Behavior

- Human behavior is probably more plastic than any other animal—as human have greater capacity for learning

Evolution and Human Culture

Control of Instinctive Behavior by Culture

- In human cultural history, humans have built up a diversity of structured societies
- Structured societies have laws, governments, institutions, cultural values and religions
- These institutions define what behavior is acceptable or what is not

Evolution and Human Culture

Control of Instinctive Behavior by Culture

- Many instinctive behaviors might enhance an individual's Darwinian fitness
- But they are considered unacceptable and prohibited by law

Evolution and Human Culture

Control of Instinctive Behavior by Culture

- These are the social and cultural institutions that define values
- Values make human behavior distinct and controlled than that of other animal societies

END